

Basel III Capital Requirements and Non-Bank Mortgage Growth: Effects on Local Housing Markets and Credit Conditions

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Abbreviations

CET1	Common Equity Tier 1
FE	Fixed Effects
FFIEC	Federal Financial Institutions Examination Council
FHA	Federal Housing Administration
FHFA	Federal Housing Finance Agency
GFC	Global Financial Crisis
GSE	Government-Sponsored Enterprise
HMDA	Home Mortgage Disclosure Act
HPI	House Price Index
HPV	House Price Volatility
HUD	U.S. Department of Housing and Urban Development
IV	Instrumental Variable
LAR	Loan Application Register
LEI	Legal Entity Identifier
LTI	Loan-to-Income
MSR	Mortgage Servicing Right
NCHS	National Center for Health Statistics
REI	Regulatory Exposure Index
RSSD	Research, Statistics, Supervision, and Discount
SAIPE	Small Area Income and Poverty Estimates
SD	Standard Deviation
SEER	Surveillance, Epidemiology, and End Results
T1RBC	Tier 1 Risk-Based Capital
USDA ERS	U.S. Department of Agriculture Economic Research Service

1 Introduction

Mortgage credit is central to both the housing market and household balance sheets. The availability and terms of mortgage credit affect housing demand and prices, driving fluctuations in the housing market. At the same time, mortgage credit shapes household balance sheets directly, by adding a non-financial asset, housing, on the assets side, and a large, long-term mortgage debt on the liabilities side. Moreover, mortgage debt is the single largest component of US household debt, around 70% of total household debt (Federal Reserve Bank of New York, 2026). Any changes in mortgage credit conditions affect household balance sheets, shaping how much leverage households carry. Highly leveraged households can amplify the effects of a macroeconomic downturn, because they are more exposed to losses in net worth following a given shock and tend to reduce their spending more sharply in response (Mian et al., 2013). This makes the composition of mortgage lending important, not just its overall availability. Therefore, this thesis aims to study the effect of the post-2013 shift in mortgage lending composition on local housing markets and credit conditions.

The mortgage market composition has been drastically changing in the United States, driven in part by the differential regulatory treatment of banks and non-bank lenders. Basel III's capital requirements, introduced in 2013, apply to depository institutions but not to non-bank mortgage lenders. Non-bank mortgage lenders, which do not take deposits, fall outside this specific capital framework, even though they remain subject to other forms of regulation.¹ Figure 1 shows the resulting pattern, a sharp rise in non-bank mortgage market share, and a corresponding fall in bank share, particularly after 2013.

A shift in who provides mortgage credit can change the conditions under which households borrow. This thesis studies the effects of this non-bank expansion on local house prices, house price volatility, borrower leverage, credit rate spreads, and housing affordability pressure, instrumenting non-bank share with each county's predetermined 2008 exposure to weak, undercapitalized banks, interacted with the timing of Basel III's 2013 implementation. This strategy isolates variation in non-bank lending associated with the regulatory-induced reallocation of mortgage credit. I further examine whether these effects vary systematically with a county's baseline minority population share, given the possibility that the reallocation of mortgage credit is not evenly distributed across communities.

The empirical evidence suggests that the rise in non-bank shares was not uniform. Counties with greater 2008 exposure to weak banks experienced a significantly larger increase in non-bank mortgage share following Basel III. This reallocation of credit towards non-banks is associated with a two-sided pattern. On one hand, it is associated with greater medium-run stability and stronger growth in house prices, consistent with non-bank lenders smoothing the local credit supply against what would otherwise have been an abrupt, regulation-driven contraction in bank

¹Non-banks remain subject to state licensing, CFPB oversight, and Dodd-Frank's Ability-to-Repay and Qualified Mortgage underwriting standards (Conference of State Bank Supervisors, 2024; United States Congress, 2010; Consumer Financial Protection Bureau, 2013)

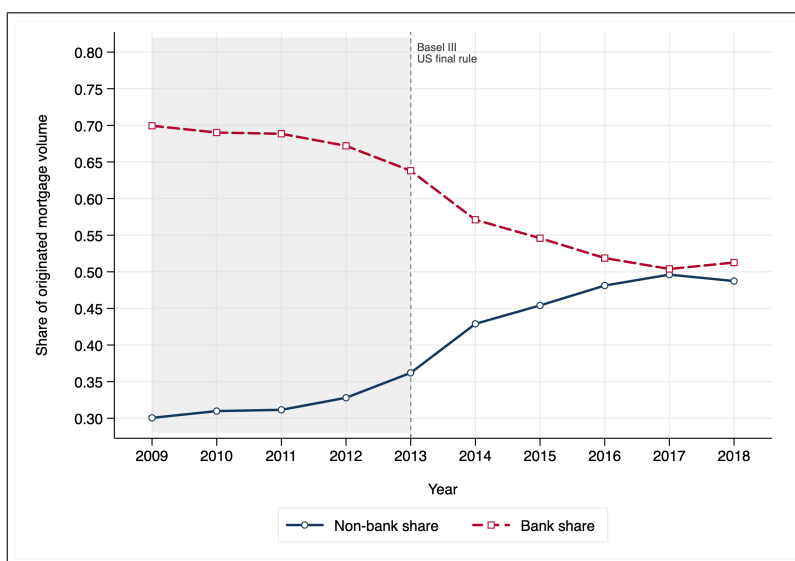
mortgage credit. On the other hand, the same reallocation is associated with rising borrower leverage, with a disproportionately larger increase in the share of high-leverage loans, wider credit rate spreads that make mortgages more expensive, and greater housing affordability pressure as house prices rise faster than household income.

These effects are not uniform across counties. The first-stage relationship and the effects on housing market outcomes are stronger in counties with higher baseline minority population shares.

Together, these findings suggest a pattern that I refer to as “stabilisation with latent fragility”. Non-bank expansion appears to have cushioned local housing markets in the medium-run while simultaneously being associated with higher leverage and credit pricing pressures.

The remainder of the thesis is structured as follows. Chapter 2 lays out Institutional Background. Chapter 3 reviews the literature. Chapter 4 describes the data. Chapter 5 presents the empirical strategy. Chapter 6 reports the main results. Chapter 7 examines the underlying mechanisms. Chapter 8 presents robustness checks. Chapter 9 discusses the results in greater depth. Chapter 10 concludes.

Figure 1. County-Level Bank and Non-Bank Mortgage Market Share, 2009–2018



Notes: Mean bank and non-bank mortgage market share by year, computed as unweighted county-level averages across 2,409 US counties. This is a broader sample than the 2,378-county main estimation sample used in the regression analysis. The vertical line marks the Basel III regulatory change. Non-bank share plus bank share equals 1 by construction.

2 Institutional Background

2.1 Basel III Capital Requirements

Basel III is an international regulatory framework developed by the Basel Committee on Banking Supervision in response to the 2007–2009 financial crisis, intended to strengthen bank capital requirements globally. In the United States, Basel III was implemented jointly by the Federal

Reserve, the Federal Deposit Insurance Corporation, and the Office of the Comptroller of the Currency, following a mandate under Section 171 of the Dodd-Frank Wall Street Reform and Consumer Protection Act (2010)(Basel Committee on Banking Supervision, 2011).

The US final rule was approved by the Federal Reserve Board on July 2, 2013, and published in the Federal Register on October 11, 2013 ². The rule set new minimum capital ratios: a Common Equity Tier 1 (CET1) ratio of 4.5%, a T1RBC ratio of 6%, up from a previous minimum of 4%, and a total capital ratio of 8%. The rule also introduced a capital conservation buffer of 2.5%, composed entirely of CET1 capital, held on top of these minimum ratios.³

2.2 The Home Mortgage Disclosure Act

The Home Mortgage Disclosure Act (HMDA) is central to this thesis, providing the loan-level data essential for the construction of bank and non-bank mortgage shares and every other loan-level outcome examined in the analysis. It was enacted by Congress in 1975 and is implemented by Regulation C ⁴. It was enacted given public concern over credit shortages in certain neighborhoods. One main purpose of HMDA is to provide the public with information that will help show whether financial institutions are serving the housing credit needs of the communities and neighborhoods in which they are located. It also helps public officials in allocating public-sector investment in ways that can encourage private investment in areas where it is needed ⁵.

2.3 Loan Types: Conforming, Non-Conforming, and FHA

The understanding of loan types for this study is crucial. As shown later in this thesis, bank and non-bank lending activity differ substantially across different loan types. This section explains the distinction between these loan types.

Conforming loans are conventional mortgages eligible for purchase by Fannie Mae or Freddie Mac, the two government-sponsored enterprises (GSEs) that dominate the US secondary mortgage market. For a loan to be eligible as a conforming loan, it must meet the GSEs' underwriting criteria and fall below an annually determined conforming loan limit. This limit is set each year by the Federal Housing Finance Agency (FHFA), as required under the Housing and Economic Recovery Act of 2008 (HERA). This limit is not the same across the nation and differs for each county; for high-cost areas, for example, higher limits are designated.⁶ A loan exceeding the applicable county limit, or otherwise failing to meet GSE underwriting standards, does not qualify as conforming.

Non-conforming loans are conventional mortgages that fail to meet the conforming loan requirements; in most cases it is because the loan amount exceeds the county's conforming limit.

²Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency (2013)

³Total Tier 1 ratio = Common Equity Tier 1 (CET1) + Additional Tier 1 ratio. Therefore, a conservation buffer of 2.5% increases the Tier 1 ratio requirement by an additional 2.5%.

⁴Consumer Financial Protection Bureau (2015), Board of Governors of the Federal Reserve System (2013)

⁵Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency (2013)

⁶FHFA's role as conservator of Fannie Mae and Freddie Mac, and its authority to set conforming loan limits, both derive from HERA (2008). For more information, see Federal Housing Finance Agency, "FHFA Conforming Loan Limit Values".

In this case the loan is described as a jumbo loan.

FHA loans are mortgages insured by the Federal Housing Administration, a US government agency, rather than purchased by a GSE. The FHA program insures loans with lower down payment and credit score requirements than conventional financing. It permits down payments as low as 3.5% for borrowers with a credit score of at least 580, and borrowers below this threshold require a down payment of at least 10% (U.S. Department of Housing and Urban Development, 2015). Unlike conforming loans, which are typically sold to Fannie Mae or Freddie Mac, FHA loans are pooled and sold into securities guaranteed by the Government National Mortgage Association, Ginnie Mae (Congressional Budget Office, 2022; Congressional Research Service, 2017).

These three categories differ not only in who holds or insures the loan, but in the degree of underwriting quality imposed on the originating lender. Details on purchased loans and purchaser types are provided in Appendix A.

3 Literature Review

A body of literature has studied the cause of the non-bank share rise after the Global Financial Crisis (GFC) and its effects on the local housing market. Building on this literature, this paper makes three contributions: identifying the effect of the 2013 Basel III minimum capital requirements on non-bank share, in counties with greater exposure to weak banks; tracing the downstream effects of this rise on local housing markets and credit conditions; examining whether these effects differ systematically across counties by minority population share.

Buchak et al. (2018) document that the market share of shadow banks in the mortgage market has nearly doubled from roughly 30% to 50% from 2007 to 2015, with the shift most pronounced in the FHA market, where shadow banks' market share increased from 45% to 75% over the same period.⁷ They find this increase is mainly attributed to regulatory changes. Using a decomposition model,⁸ they estimate that increasing regulatory burden faced by traditional banks created an arbitrage for the non-banks. In addition, they also find financial technology to be another aspect that accounts for shadow bank growth.

However, Buchak et al.'s regulatory-pressure measures, bank capital changes, and mortgage servicing rights treatment are estimated as a single cross-sectional long difference between 2008 and 2015. As a first stage, this paper builds an exposure index at county level which shows differential effects of the regulatory changes. In other words, this paper shows the non-bank share has not increased uniformly across all counties; rather, counties with greater 2008 exposure to weak banks experienced a significantly larger increase in non-bank share following the 2013 Basel III regulatory changes. To address this, this paper uses a shift-share-based instrument, interacting a predetermined 2008 exposure index with the 2013 regulatory shock.

⁷The non-bank mortgage lenders considered in this thesis are closely related to what the literature commonly refers to as shadow banks, although the two terms are not strictly equivalent.

⁸Buchak et al. (2018) decompose shadow bank growth using a structural discrete choice model, attributing roughly equal shares to regulatory burden on traditional banks and technological improvements by shadow bank lenders. This paper isolates the regulatory channel through a shift-share instrumental variables strategy.

Nevertheless, this paper draws on Buchak et al.’s mechanism to explain the effect of capital requirement changes on non-bank shares. Their paper is an anchor for this paper’s first stage, and the choice of the Tier 1 risk-based capital ratio as the regulatory measure is inspired by theirs.

Choi et al. (2024), another related paper this thesis builds on, study the effect of uneven non-bank mortgage expansion on localized housing market dynamics. They instrument non-bank credit supply with the local conforming loan-eligible share, and find that increased non-bank lending drives localized housing booms, characterized by rapid home price appreciation, rising transaction volumes, and intensified market competition. They also find these effects are especially persistent in census tracts near affluent neighborhoods.

In sum, they study the effect of non-bank share on similar outcomes, however, instrumented with the conforming loan-eligible share. This paper specifically looks at the effects of the non-bank share rise driven by the 2013 regulatory changes on local housing conditions like annual house price growth, house price volatility, and three-year cumulative growth, along with some additional housing stability measures.⁹ It further extends this analysis to credit conditions and housing affordability pressure.

Another related work by Mian et al. (2013) finds that ZIP codes with poorer and more highly leveraged households have a significantly higher marginal propensity to consume out of housing wealth. Their findings suggest that highly leveraged households are more responsive to changes in housing wealth and experience larger reductions in consumption during downturns.

This finding is directly relevant to this thesis’s own results. This thesis documents a rise in average borrower loan-to-income ratio, and a substantially larger rise in the share of loans exceeding a high-leverage threshold, following non-bank expansion (Section 7.3). Taking Mian et al. (2013) as motivation, this thesis interprets the rise in leverage as a plausible source of latent fragility. Counties with greater non-bank-driven leverage growth may be more exposed to future wealth or income shocks, since more highly leveraged households respond more sharply to such shocks.

This thesis contributes to the literature in three ways. First, it isolates the effect of Basel III capital requirements on the share of mortgage lending supplied by non-banks using a shift-share instrumental variables strategy, rather than examining regulatory pressure more broadly without a comparable identification strategy in Buchak et al. (2018). Second, it traces the effects of this reallocation beyond house prices to local housing-market volatility, borrower leverage, credit rate spreads, and housing affordability pressure, extending the focus on house prices in Choi et al. (2024). Third, it examines whether the effects differ across counties according to their baseline

⁹Additional housing stability measures such as price growth persistence and peak-to-trough drawdown.

minority population share, a dimension that has received little direct attention in this literature.

4 Data

The estimation panel is at the county-year level, running from 2009 to 2018. The main estimation sample contains 23,130 observations across 2,378 counties.¹⁰ Sample size varies slightly across tables depending on each outcome’s own coverage, and this is noted where relevant.

4.1 Data Sources

The panel combines loan-level mortgage data, bank balance sheet data, house price data, and county-level demographic and economic characteristics, described in turn below. The panel is constructed from eight data sources.

Loan-level mortgage data come from the Home Mortgage Disclosure Act (HMDA) Loan Application Register (LAR), covering 2008 through 2018.¹¹ I use the nationwide first-lien, owner-occupied, one-to-four-family loan records. Bank balance sheet data come from FFIEC Call Reports, specifically the Tier 1 risk-based capital ratio reported on Schedule RC-R. House price data come from the Federal Housing Finance Agency (FHFA) House Price Index, at the county level, a comprehensive collection of publicly available annual house price indexes measuring changes in single-family home values.¹²

Median household income and poverty rates are obtained from the Census Bureau’s Small Area Income and Poverty Estimates (SAIPE) program. County-level unemployment rates come from the United States Department of Agriculture Economic Research Service (USDA ERS). Population data are obtained from the Census Bureau. Minority population shares are constructed separately using bridged-race postcensal population estimates produced by the Census Bureau in collaboration with the National Center for Health Statistics (NCHS) and distributed through the Surveillance, Epidemiology, and End Results (SEER) Program of the National Cancer Institute.

Residential building permit data are obtained from the United States Department of Housing and Urban Development (HUD) GIS Helpdesk.

Finally, linking HMDA respondent identifiers to bank balance sheet data requires a crosswalk between HMDA’s respondent identification system and the Federal Reserve’s RSSD identifiers. I use the HMDA lender crosswalk, commonly referred to as the Avery file, originally constructed by Robert Avery and distributed by Neil Bhutta through the Federal Reserve Bank of Philadelphia. A full account of all data sources is reported in Appendix B.

¹⁰This main sample is obtained after standard listwise deletion and the exclusion of singleton counties required for fixed-effects estimation.

¹¹The 2018 file follows a different HMDA reporting structure, introduced under the 2015 HMDA rule. However, it follows the same filters as pre-2018.

¹²The FHFA HPI is constructed using a weighted repeat-sales methodology.

4.2 Data Construction

4.2.1 Bank and Non-Bank Shares

The lender status is not observed directly in the HMDA loan-level file. The unique identifier for the institutions is the concatenation of Agency Code and Respondent ID for 2008 to 2017, and Agency Code and LEI (Legal Entity Identifier) for 2018. However, to classify a lender as a traditional bank or a non-bank, the nature of the lender must be known. For this, I use a separate HMDA auxiliary file, the Reporter Panel, which characterizes each filing institution rather than each individual loan. The `other_lender_code` field in the Reporter Panel identifies depository institutions directly, and plays a key role in lender classification.

I define a bank as a depository institution. A financial institution legally permitted to accept consumer deposits, including commercial banks, savings institutions, and credit unions. Any non-depository institution is defined as a non-bank, which predominantly includes independent mortgage banking subsidiaries.

The loan-level file is merged with the Reporter Panel using these unique identifiers, and bank and non-bank shares are then calculated, which are used intensively throughout the study.

Data cleaning. For variables with outliers in the sample, I use winsorization to filter the data. Observations falling above the 99th percentile and below the 1st percentile are excluded.

4.2.2 Housing Outcomes

Using county-level FHFA House Price Index data for 2005–2018, indexed to a 2000 base year (`hpi_2000`), I construct six housing market outcomes: annual house price growth, house price volatility, three-year cumulative house price growth, price growth persistence, peak-to-trough drawdown, and the log price-to-income ratio. The house price series is extended back to 2005, four years before the first year of the estimation panel (2009), to ensure that the rolling-window measures have sufficient historical observations and are fully defined from the beginning of the estimation period.

House price volatility. Three-year house price volatility, `hpi_3yr`, is the rolling standard deviation of `hpi_log_growth` over a three-year window, covering the current year and the two preceding years:

$$\text{hpi_3yr}_{ct} = \text{SD}\left(\text{hpi_log_growth}_{c,t-2}, \text{hpi_log_growth}_{c,t-1}, \text{hpi_log_growth}_{ct}\right)$$

This construction captures how much annual house price growth fluctuates within a local housing market over a relatively short period. A higher value therefore indicates less stable year-to-year price growth, while a lower value indicates that annual growth is more closely clustered around its local mean. The rationale for a three-year window and construction of other housing outcomes is described in detail in [Appendix B](#).

4.2.3 Credit Rate Spreads and Higher-Priced Mortgage Lending

To examine mortgage pricing following the reallocation toward non-bank lenders, I construct two county-year measures from the HMDA loan-level data: the average credit rate spread among higher-priced mortgages, and the share of mortgage lending classified as higher-priced. Both measures are constructed using loan amount as the weighting variable, so that larger mortgages receive proportionately greater weight in the county-level measures.

For 2009–2017, HMDA reports the rate spread only for loans that meet the applicable threshold for higher-priced mortgage reporting. I therefore use the reported rate spread for these observations. For 2018, the HMDA reporting framework changed, and I apply a 1.5 percentage-point threshold explicitly to identify higher-priced loans. The study sample therefore contains rate spread observations at or above the 1.5 percentage-point threshold.

Construction of average rate spread, higher-priced loan share, and other borrower leverage outcomes like average Loan-to-Income ratio and high Loan-to-Income Ratios are reported in Appendix B.

4.2.4 Minority Population Share

The data I use provide county-level population counts by race, Hispanic origin, sex, and age for each year. I use these data to construct county-year measures of minority population composition for 2009–2018.

The primary measure defines the minority population as the proportion of the county population that is not non-Hispanic White. This definition follows the minority population classification in Crosignani and Le (2023). I then divide it by total population to calculate the minority share.¹³ A detailed construction of this county-level minority share is described in Appendix B.

4.3 Descriptive Statistics

Table 1 reports descriptive statistics for the main estimation sample used in the first-stage and outcome regressions, computed at the county-year level over 2009–2018.

Non-bank mortgage share averages 39.6% across the sample, with substantial variation across counties (standard deviation 16.0 percentage points, ranging from 1.2% to 93.1%). Regulatory exposure averages 0.279, with a wide range from 0 to 1, reflecting the underlying variation in 2008 weak-bank presence across counties.

The housing market variables also show considerable variation. Annual log house price growth averages 0.84%, with a standard deviation of 5.10 percentage points. The corresponding three-year cumulative house price growth averages 1.38%, with a standard deviation of 11.7 percentage points. Three-year house price volatility has a mean of 0.0287 and a standard deviation of 0.0227. The relatively wide range of the growth measures reflects substantial differences in local house price dynamics across counties and years.

¹³They define minority population as total Hispanic population plus non-Hispanic population minus non-Hispanic White alone population. This is equivalent to one minus the non-Hispanic White share of total population.

Average borrower leverage, measured by the loan-to-income ratio, is 2.24, with a standard deviation of 0.43. The average rate spread on higher-priced loans is 2.37 percentage points, with a standard deviation of 0.69 percentage points. The minimum is 1.50 percentage points, reflecting the threshold used to identify higher-priced loans in the data. This variable has a smaller sample ($N = 23,047$) than the rest of the table, because it is undefined for county-years with no loans meeting the higher-priced threshold.

For the variables that capture local economic and housing-market conditions, the average county unemployment rate is 6.82%, while median household income averages approximately \$49,193. Minority population share averages 21.2%, with substantial cross-county variation, ranging from approximately 1.1% to 97.2%.

Overall, the descriptive statistics show substantial variation in mortgage market structure, regulatory exposure, housing outcomes, and local economic conditions across the county-year observations used in the analysis. This variation, together with the pre-treatment differences noted above, motivates the empirical strategy discussed in Chapter 5. Additional descriptive statistics including the variables used in robustness checks are described in Appendix Table 15.

Table 1. Summary Statistics: Main Outcome Regressions

	Count	Mean	SD	Min	Max
Non-bank mortgage share	23,130	0.396	0.160	0.012	0.931
Regulatory Exposure Index	23,130	0.279	0.169	0.000	1.000
House price volatility (3yr)	23,130	0.029	0.023	0.002	0.132
Annual house price growth	23,130	0.008	0.051	-0.426	0.350
3yr cumulative house price growth	23,130	0.014	0.117	-0.377	0.291
Average loan-to-income ratio	23,130	2.243	0.428	1.064	4.292
Average LTI (lagged)	23,130	2.231	0.426	1.064	4.292
Average credit rate spread	23,047	2.370	0.686	1.500	4.999
Log price-to-income ratio	23,130	-5.857	0.244	-6.499	-5.322
Unemployment rate (%)	23,130	6.821	2.957	1.100	29.400
Median household income (\$000s)	23,130	49.193	12.756	23.014	140.382
Minority share	23,130	0.212	0.183	0.011	0.972
Log HPI (lagged)	23,130	4.908	0.165	4.132	5.971
Population growth	23,130	0.003	0.011	-0.184	0.211
Log permits (lagged)	23,130	4.122	1.681	0.693	9.827
Poverty rate (%)	23,130	15.444	5.604	2.600	45.000
Baseline minority share (2008)	23,130	0.197	0.179	0.011	0.970

Notes: Summary statistics for the main estimation sample ($N = 23,130$ county-year observations, 2,378 counties, 2009–2018). Covers variables used in the housing market, credit condition, affordability, and minority-share heterogeneity regressions. Poverty rate and baseline minority share are used only in the affordability and heterogeneity specifications respectively, and are not part of the main first-stage control set. Non-bank mortgage share is volume-weighted. The log price-to-income ratio is $\log(\text{house price index}) - \log(\text{median household income})$, so its level reflects the differing units of the two series and is not directly interpretable; only changes in it are meaningful.

5 Empirical Strategy

5.1 Identification Challenge

The core problem in studying the effect of non-bank mortgage market share on housing outcomes and financial fragility measures is the endogeneity of non-bank share. Non-banks are not

randomly assigned across counties. They actively choose where to operate based on borrower composition (Dolson and Jagtiani, 2021; Gete and Reher, 2021) and profit opportunities created by bank retreat (Sarto and Wang, 2026). For instance, counties with more non-bank activity may have already had more stable or appreciating housing markets before non-banks arrived, or non-banks may have disproportionately entered counties with pre-existing riskier borrower pools. In any of these cases, OLS would attribute to non-bank entry outcomes that were already present, or already improving, before non-banks arrived, and cannot distinguish the causal effect of non-bank lending from the pre-existing conditions that attracted non-banks in the first place. OLS estimates of the effect of non-bank share on housing outcomes would therefore be biased, plausibly toward overstating the causal effects found here.

A further source of endogeneity arises from the regulatory environment itself. Non-bank mortgage market share responds endogenously to regulatory pressure on banks. Buchak et al. (2018) show that shadow banks entered the markets where traditional banks contracted due to regulatory constraints, implying regulatory shocks to banks predict non-bank mortgage shares. Without isolating the regulatory component of non-bank expansion, OLS conflates its effect on housing outcomes with regulatory factors affecting both lending and local markets. Hence, an instrument is needed that predicts non-bank share through a channel unrelated to the pre-existing county characteristics and trends that drive non-bank location decisions. The next section outlines the identification strategy, which combines two-way fixed effects with an instrumental variables approach to address these concerns.

5.2 Identification Strategy

5.2.1 Two-Way Fixed Effects

I address the first layer of endogeneity through a two-way fixed effects specification that absorbs county and year fixed effects in all regressions. County FE absorbs all time-invariant county characteristics; this includes the permanent structural features of local housing markets, the long-run demographic and income composition of borrower pools, and the historical concentration of bank types in each county. Year fixed effects absorb common national trends, including national monetary policy movements, fintech growth, and any other shocks that affect all counties simultaneously. Together, county and year fixed effects ensure that all estimates are identified from within-county variation over time, after removing permanent county characteristics and common national trends. However, TWFE does not address time-varying within-county endogeneity. Hence, an instrumental variables strategy is also used for the estimation.

5.2.2 Construction of the shift-share-based Instrument

This section describes the construction of the instrument using the Regulatory Exposure Index (hereafter REI), the main measure of county-level exposure to weak banks in 2008. This variation is central because not all counties were affected uniformly by the Basel III regulatory changes.

Motivation and logic. 2008 is chosen as the base year for county exposure for two main reasons. First, 2008 predates the Basel III announcement by at least two years, ensuring the

exposure measure is fully predetermined with respect to the regulatory shock: banks could not have adjusted their 2008 capital position in anticipation of regulations that did not yet exist. Second, 2008 represents the earliest year for which there is comprehensive HMDA loan-level data, providing reliable county-level lending shares for constructing the exposure measure.

In addition, weak banks are defined as banks that maintained less than 8% of the Tier 1 risk-based capital ratio (hereafter T1RBC ratio) in 2008. Any bank with less than 8% T1RBC ratio is either considered as not maintaining enough capital or has thin capital buffers. Although Basel III sets the minimum T1RBC ratio at 6%, the fully phased-in capital conservation buffer raises the effective requirement to 8.5%. However, this buffer was phased in only from 2016 and was not fully binding until 2019, after the end of the sample period. I retain the 8% threshold as the main specification because it produces the strongest first-stage relationship among the thresholds considered, with an excluded-instrument F-statistic of 48.95, compared with 47.44 under the 8.5% threshold. The results are robust to using an 8.5% threshold or a continuous exposure measure defined as the county-level weighted average of the inverse T1RBC; both produce positive and statistically significant first-stage estimates (Section 8.1).

The T1RBC ratio is used because it is the primary regulatory target of Basel III, and because the denominator of this ratio is risk-weighted assets. Residential mortgages carry risk weights of 50% to 100% under the standardised approach (Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency, 2013), meaning every additional mortgage originated increases risk-weighted assets and mechanically reduces the T1RBC ratio. Banks facing binding capital constraints therefore had a direct incentive to cut mortgage lending specifically. Hence, counties highly exposed to weak banks saw bank mortgage lending fall, creating a gap that less-regulated non-bank lenders moved to fill, leading to differential local housing market outcomes and credit conditions.¹⁴ I refer to this as the reallocation mechanism throughout the thesis.

Construction of the shift and share components. I construct a shift-share-based instrument following the tradition of Bartik (1991) and Goldsmith-Pinkham et al. (2020), exploiting the interaction between predetermined county-level exposure to weak banks and the common national regulatory shock of Basel III implementation.¹⁵

The share component. The share component measures each county’s 2008 exposure to weak banks: specifically, the fraction of county mortgage originations from banks with a T1RBC ratio below 8% in 2008. First, for each bank operating in a county, I compute its share of the county’s total mortgage origination volume in 2008:

$$\text{Share}_{bank,c} = \frac{\text{Loans}_{bank,c,2008}}{\text{Total bank loans}_{c,2008}},$$

¹⁴This mechanism follows Buchak et al. (2018), who show banks facing greater regulatory burden retreated from mortgage lending, with shadow banks partially filling the gap.

¹⁵The instrument follows the shift-share logic of the Bartik literature, combining predetermined local exposure with a common aggregate shock, but it differs from the standard Bartik instrument in that the time-varying component is a single regulatory implementation shock rather than a sequence of industry-specific national shocks.

where $\text{loans}_{\text{bank},c,2008}$ is the volume of mortgage loans originated by each bank in county c in 2008, and $\text{total_bank_loans}_{c,2008}$ is the total county bank mortgage origination volume in the same year.

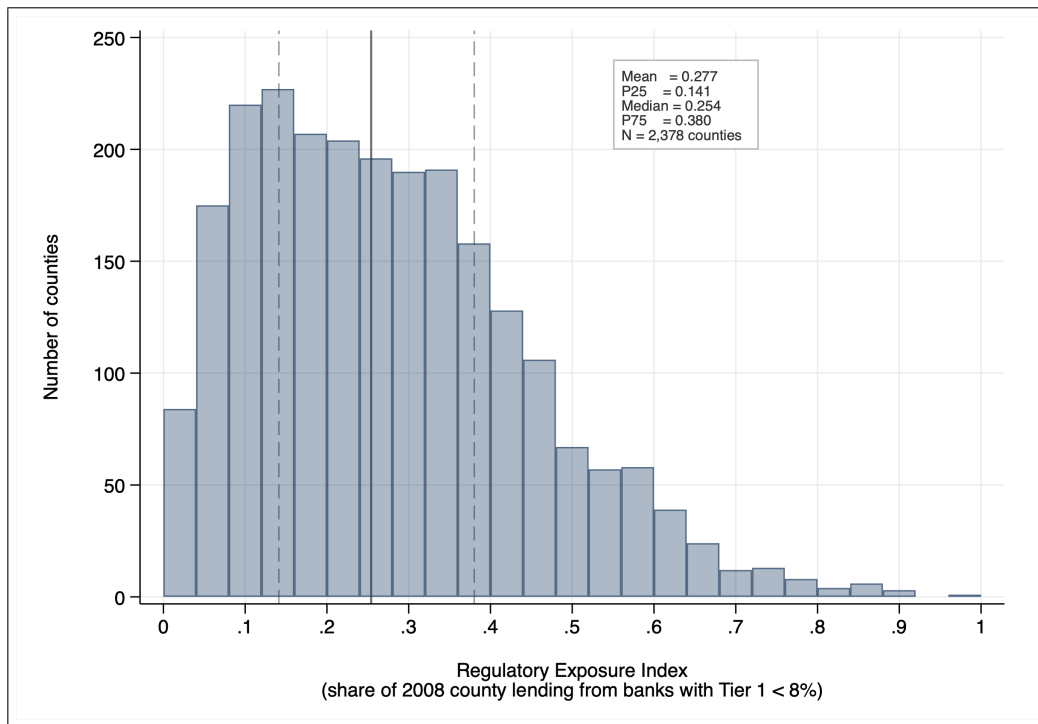
Using the indicator $\mathbf{1}[\text{Tier1}_{\text{bank},2008} < 0.08]$, which takes the value one if a bank's T1RBC ratio fell below 8% in 2008 and zero otherwise, the Regulatory Exposure Index sums only the lending shares of weak banks:

$$\text{REI}_c = \sum_{\text{bank}} \text{Share}_{\text{bank},c} \cdot \mathbf{1}[\text{Tier1}_{\text{bank},2008} < 0.08].$$

To illustrate, a county where weak banks originated 60% of all bank mortgages in 2008 receives an exposure index of 0.60, while a county where weak banks originated only 10% receives an index of 0.10. The index therefore captures how much of each county's 2008 mortgage market was concentrated among banks that would subsequently face binding Basel III capital pressure.

Figure 2 shows the distribution of the Regulatory Exposure Index across 2,378 counties. The index ranges from 0 to 1, with a mean of 0.277 and a median of 0.254, indicating that the typical county had roughly a quarter of its 2008 mortgage market served by weak banks. The distribution is right-skewed; most counties fall between the 25th percentile (0.141) and the 75th percentile (0.380), while a smaller number of counties have substantially higher exposure. This cross-county variation in 2008 exposure to weak banks forms the basis of the identification strategy.

Figure 2. Distribution of the Regulatory Exposure Index Across Counties



Notes: Distribution of the Regulatory Exposure Index across 2,378 counties in the main estimation sample. The solid vertical line marks the median; dashed vertical lines mark the 25th and 75th percentiles. Index measures the share of 2008 county mortgage origination volume from banks with Tier 1 risk-based capital ratios below 8%, constructed from HMDA LAR and FFIEC Call Report data.

The shift component. The shift component captures the timing of the regulatory shock. A binary post indicator equals one from 2013 onward:

$$\text{Post}_t = \mathbf{1}[\text{year} \geq 2013].$$

The Board of Governors approved the US Basel III final rule in July 2013, with publication in the Federal Register on October 11, 2013 (Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency, 2013). This regulatory change triggered active balance sheet adjustment for weak banks.

The shift component is identical across all counties: every county faces the same post-2013 regulatory environment. All cross-county variation in the instrument therefore comes entirely from the predetermined share component.

The instrument. The instrument is the product of the two components:

$$Z_{ct} = \text{REI}_c \times \text{Post}_t.$$

Before 2013, the instrument equals zero for all counties: high- and low-exposure counties faced the same regulatory environment. After 2013, counties with higher weak bank exposure faced greater regulatory pressure on their local banks, triggering the reallocation mechanism. The instrument generates variation in non-bank expansion driven entirely by the predetermined composition of each county’s 2008 bank market, not by post-2013 housing market conditions or non-bank location decisions.

This paper exploits the shift-share-based instrument constructed here to study the effect of non-bank share on local housing outcomes and credit conditions.

5.3 Estimation

The empirical strategy proceeds in two stages. The first stage estimates the effect of regulatory exposure on non-bank mortgage market share:

$$\text{Nonbank_share}_{ct} = \alpha + \beta(\text{REI}_c \times \text{Post}_t) + X'_{ct}\gamma + \delta_c + \delta_t + \varepsilon_{ct}, \quad (5.1)$$

where $\text{Nonbank_share}_{ct}$ is the non-bank share of total mortgage origination volume in county c in year t , $\text{REI}_c \times \text{Post}_t$ is the Regulatory Exposure Index interacted with the post-2013 indicator, X'_{ct} is a vector of time-varying county controls, δ_c are county fixed effects, δ_t are year fixed effects, and ε_{ct} is an idiosyncratic error term.¹⁶ The coefficient $\hat{\beta}$ captures the differential growth in non-bank share in counties with greater 2008 exposure to weak banks following Basel III implementation.

The second stage uses the predicted values from the first stage to estimate the causal effect of

¹⁶The vector of controls X'_{ct} varies across specifications depending on the outcome variable. The main specification includes lagged average borrower LTI, unemployment rate, lagged log HPI, population growth, median household income, minority share, and lagged log permits.

non-bank mortgage expansion on each outcome variable:

$$Y_{ct} = \alpha + \rho \cdot \widehat{\text{Nonbank_share}}_{ct} + X'_{ct}\gamma + \delta_c + \delta_t + \varepsilon_{ct}, \quad (5.2)$$

where Y_{ct} is one of the local housing market outcomes or credit conditions, including three-year house price volatility, house price growth, average borrower loan-to-income ratio, average credit rate spread, and the log price-to-income ratio in county c and year t , and $\widehat{\text{nonbank_share}}_{ct}$ are the fitted values from the first stage. The IV estimator recovers the Local Average Treatment Effect (LATE), the causal effect of non-bank mortgage expansion on local housing markets and credit conditions for the subset of counties where non-bank share increased specifically because of Basel III capital pressure on weak banks.¹⁷ The coefficient $\hat{\rho}$ captures the causal effect of a one percentage point increase in non-bank mortgage market share on the outcome variables.

To examine whether the first-stage relationship and the housing-market effects of non-bank mortgage expansion differ with counties' baseline minority share, I extend the baseline IV specification by interacting both the regulatory exposure measure and non-bank mortgage share with baseline minority share. The estimation equations and the coefficient interpretations are provided in Appendix C.

5.4 Identification Assumptions

The coefficient $\hat{\rho}$ on non-bank share has a causal interpretation under three conditions: the instrument is relevant, the exclusion restriction holds, and the instrument satisfies monotonicity. Each condition is discussed in this section.

5.4.1 Relevance

The relevance assumption requires that the instrument, REI interacted with the post-2013 indicator, is a strong predictor of county-level non-bank mortgage market share after conditioning on controls and fixed effects.¹⁸

Following the reallocation mechanism described in Section 5.2.2, the first-stage results in Table 4 confirm the strong relationship between the instrument and non-bank share. In the standardized specification, the estimated coefficient is 1.22 and statistically significant at the 1% level. This implies a one standard deviation increase in regulatory exposure raises non-bank mortgage share by 1.22 percentage points after 2013, corresponding to approximately 3.1% of the sample mean non-bank share. The first-stage F-statistic of 48.95 is well above the Stock and Yogo (2005) weak-instrument threshold of 16.38, providing evidence against weak instrument concerns.¹⁹

The event study in Figure 3 confirms the effect is concentrated post-2013, with β coefficients on the instrument close to zero before 2013 and rising sharply from +0.0262*** in 2013 to

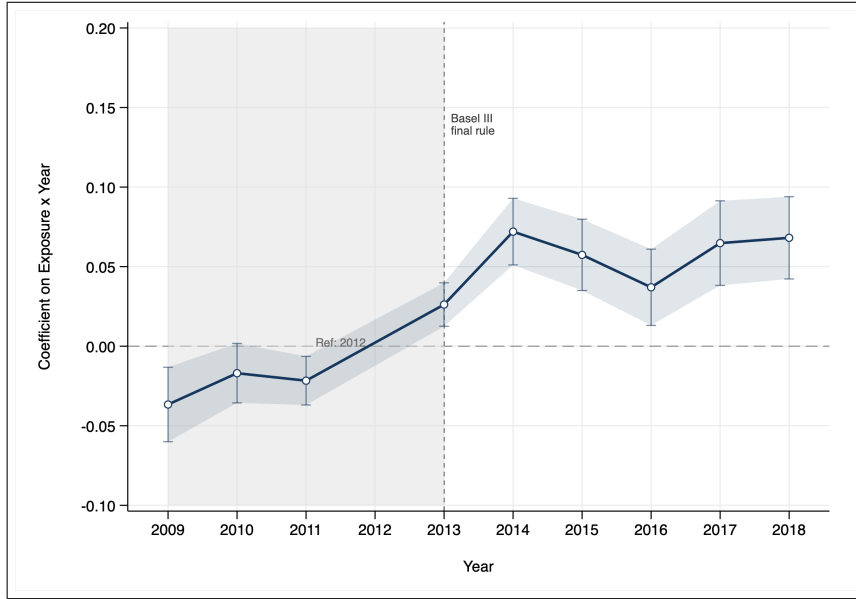
¹⁷This is the most policy-relevant parameter of the research question, as it directly captures the housing market consequences of regulation-induced mortgage market restructuring.

¹⁸Formally: $\text{Cov}(\text{REI}_c \times \text{Post}_t, \text{nonbank_share}_{ct}) \neq 0$.

¹⁹The F-statistic is the cluster-robust Wald test on the excluded instrument (exposure_post), numerically equivalent to the Kleibergen–Paap rk Wald F-statistic in this just-identified setting (Stock and Yogo, 2005).

+0.0720*** in 2014 after Basel III capital requirements became binding. ²⁰

Figure 3. Event Study: Coefficients on Regulatory Exposure by Year Relative to 2012



Notes: Coefficients from regressing non-bank share on county exposure $\times$ year indicators, with 2012 as the omitted reference year (not shown). Shaded area shows the 95% confidence interval. Shaded background marks the pre-Basel III period (2009–2012); vertical line marks 2013, the year of Basel III’s final rule. Pre-period coefficients (2009–2011) are consistently negative and jointly significant ($F(3, 2377) = 4.37, p = 0.0045$); Controls: lagged average LTI, unemployment rate, lagged log HPI, population growth, median household income, minority share, lagged log permits. County and year fixed effects absorbed. Standard errors clustered at the county level.

5.4.2 Instrument Validity

A major threat is the effect of the GFC on banks’ capital ratios pre-2013. Siedlarek (2024) establishes that banks raised T1RBC ratios following the Global Financial Crisis independently of Basel III, as market pressure and supervisory guidance incentivised capital rebuilding throughout 2009–2012. If this post-GFC adjustment affected mortgage lending differentially in high-exposure counties, the instrument could capture the GFC channel rather than the Basel III regulatory channel. I examine pre-existing differential trends to test whether this mechanism poses a threat to identification.

Pre-Existing Differential Trends

The event study results reveal statistically significant pre-treatment interaction coefficients, and a joint F-test rejects the null that all pre-treatment coefficients are jointly equal to zero ($F(3, 2377) = 4.37, p = 0.0045$). This indicates that high- and low-exposure counties followed somewhat different trajectories in non-bank mortgage market share prior to the implementation of Basel III. However, three pieces of evidence suggest that these differential trends are unlikely to invalidate the identification strategy.

²⁰Look at Table 18 in Appendix C for the event study table with coefficients for each year.

First, from Table 18 in Appendix C, the estimated pre-treatment coefficients are consistently negative (-0.0366^{***} , -0.0170^* , and -0.0217^{***} for 2009–2011), implying that, relative to the 2012 reference year, high-exposure counties exhibited lower non-bank mortgage market shares before 2013. This pattern runs in the opposite direction from what the GFC-driven capital-rebuilding channel would predict. If differential post-GFC adjustment were driving the result, high-exposure counties should have shown higher rather than lower non-bank share growth during 2009–2011. It is also inconsistent with the concern that counties with higher exposure were already experiencing faster growth in non-bank mortgage lending prior to the reform. This evidence is further supported by a placebo test restricting the sample to 2009–2012 and assigning a fake treatment period in 2011–2012; the estimated placebo effect is economically small and statistically insignificant ($\beta = 0.00134$, $p \approx 0.87$).

Second, Table 2 shows the first-stage relationship remaining positive and highly statistically significant after adding county-specific linear time trends (0.052^{***} , $F = 31.28$), which directly absorbs pre-existing differences across counties. If the result were driven by pre-existing differential trends, absorbing those trends should render the coefficient insignificant, but it does not.

Third, Roth (2022) argues that statistically significant pre-treatment coefficients should not automatically be interpreted as evidence against causal identification. Rather, researchers should assess whether the observed pre-trends are economically capable of explaining the estimated treatment effects, and examine the robustness of the results to alternative specifications. In this study, the post-treatment effects (0.0262 – 0.0720) are substantially larger than the pre-treatment coefficients (0.0170 – 0.0366 in absolute terms), and the result remains robust after controlling for county-specific linear time trends, satisfying both conditions identified by Roth (2022).

Table 2. County-Specific Linear Time Trends

	(1) Main	(2) County Trends
Regulatory Exposure Index $\times$ Post	0.072^{***} (0.010)	0.052^{***} (0.009)
Observations	23,130	23,130
Within R-squared	0.046	0.005
F-statistic	48.95	31.28

Notes: Column (2) adds county-specific linear time trends (`county_id#c.year`) to the main specification, absorbing any smooth, county-specific pre-existing trajectory in non-bank share. A coefficient surviving this stricter specification is evidence against pre-existing differential trends driving the main result, though this does not address differences in the pre-treatment level of county characteristics. All specifications include county and year fixed effects and the control set described in Appendix C. Standard errors, clustered at the county level, are in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4.3 Exclusion Restriction

The exclusion restriction requires that the Regulatory Exposure Index affects housing market outcomes and credit conditions solely through its effect on non-bank mortgage market share, and

not directly or through any other channel that bypasses non-bank share. I provide theoretical justifications and two empirical tests in support of this assumption.

REL_c is constructed using banks' T1RBC ratios and county-bank mortgage lending shares observed in 2008, predetermined with respect to Basel III. Conditional on the included controls and county and year fixed effects, any remaining effect of the instrument on housing outcomes and credit conditions is therefore assumed to operate exclusively through non-bank share. The control variables capture local housing demand, macroeconomic conditions, and housing supply, thereby reducing the possibility that regulatory exposure affects housing outcomes through broader local economic channels.

A potential threat to the exclusion restriction is that weak banks' cutback in lending due to capital restrictions could affect housing and credit market activity directly, through a general credit contraction, rather than only through the non-bank reallocation. Table 3 presents reduced-form estimates of the effect of the Regulatory Exposure Index on lending volumes. Bank mortgage origination volume fell by 11.8% in high-exposure counties post-2013 (column 2), consistent with weak banks cutting mortgage lending to raise their T1RBC ratios. Non-bank origination volume rose by 17.5% (column 1), more than offsetting the bank retreat, so total mortgage origination volume expanded by 5.1% (column 3). This is consistent with the effects documented in Chapter 6 operating through the reallocation from bank to non-bank lending rather than through a general contraction in total mortgage credit.

Table 3. Effect of Regulatory Exposure on Lending Volume

	(1)	(2)	(3)
	Log NB Volume	Log Bank Volume	Log Total Volume
Regulatory Exposure Index $\times$ Post	0.161*** (0.037)	-0.125*** (0.030)	0.049** (0.024)
Observations	23,130	23,130	23,130
Exact % change	17.5	-11.8	5.1

Notes: Reduced-form estimates of the effect of regulatory exposure on log lending volume, by lender type and overall. All specifications include county and year fixed effects and the control set described in Appendix C, excluding lagged average LTI, conforming share, and FHA share, since these are themselves outcomes examined elsewhere in this thesis and not appropriate as controls here. Standard errors, clustered at the county level, are in parentheses. Exact percent change is computed as $100(e^{\hat{\beta}} - 1)$, the appropriate transformation for a log-linear outcome. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A potential alternative channel exists through bank lending standards. If weak banks in high-exposure counties cut lending by selectively rejecting high-LTI borrowers, the instrument could directly reduce county-level average LTI and credit rate spread without operating through the non-bank share channel. To test this, I run reduced-form regressions of bank average LTI and bank high-LTI share on the instrument. If a tightening of lending standards were the operative channel, both outcomes should fall in high-exposure counties post-2013 as banks shed riskier borrowers. Instead, both coefficients are positive and statistically significant: bank average LTI rose by 7.66 percentage points, and bank high-LTI share rose by 1.53 percentage points in high-exposure counties post-2013 (Table 16). This rules out bank lending-standards tightening

as an alternative channel.

5.4.4 Monotonicity

The monotonicity assumption requires that higher regulatory exposure weakly increases non-bank mortgage share for all counties. The tercile-split specification (Appendix Table 17) provides direct empirical support. Relative to low-exposure counties (T1), medium-exposure counties (T2) experienced 1.33 percentage points more non-bank share growth post-2013 ($p < 0.001$), and high-exposure counties (T3) experienced 3.09 percentage points more ($p < 0.001$). A formal test of equality between the T3 and T2 coefficients strongly rejects the null ($F(1, 2377) = 22.40$, $p < 0.0001$), confirming that the first-stage effect scales continuously with regulatory exposure intensity rather than reflecting a binary treatment effect. Although monotonicity cannot be tested directly, these results are consistent with the assumption.

6 Empirical Results

6.1 First Stage: Regulatory Exposure and Non-Bank Credit Growth

Table 4 presents the first-stage relationship between the Regulatory Exposure Index interacted with the post-2013 indicator and county-level non-bank mortgage market share. In Column (1), the coefficient on $REI_c \times Post_t$ is 0.0724, significant at the 1% level, with an F-statistic of 48.95. In standardized terms, a one standard deviation increase in regulatory exposure in 2008 raises non-bank share by 1.22 percentage points after 2013. Since the average non-bank mortgage share is 39.6%, this corresponds to an increase of approximately 3.1% relative to the sample average.

Table 4 presents the first-stage estimates across three specifications. Column (1) is the main specification, using the county and year fixed effects and controls described in Appendix C. Columns (2) and (3) add lagged and contemporaneous mortgage composition controls as robustness checks. The coefficient on $REI \times Post$ ranges from 0.0724 to 0.0894 across specifications, all significant at the 1% level. Further, interpretation of the tercile split specification is discussed in Appendix Table 17.

The event study in Figure 3 shows this response is not uniform across all years post-2013. The effect peaks in 2014 at 7.20 percentage points and settles into a lower but still elevated range of 3.70 to 6.81 percentage points through 2018. This pattern is consistent with weak banks adjusting most sharply in the immediate aftermath of the Basel III final rule, with non-bank share rising alongside this initial adjustment before stabilizing at a persistently higher level.

This shift reflects a reallocation of mortgage credit among lenders rather than a general contraction. Section 5.4.3 shows non-bank origination volume rising by more than bank volume falls, so total credit expands in high-exposure counties post-2013. Chapter 7 examines the channels behind this reallocation in detail.

Table 4. First-Stage Estimates: Effect of Regulatory Exposure on Non-Bank Mortgage Market Share

	<i>Dependent variable: non-bank mortgage market share</i>		
	(1) Main	(2) Lagged	(3) Contemporaneous
<i>Panel A: Raw coefficient (main specification)</i>			
REI $\times$ Post	0.072*** (0.010)	0.088*** (0.010)	0.089*** (0.010)
Unemployment rate	−0.007*** (0.001)	−0.006*** (0.001)	−0.006*** (0.001)
Log HPI (lagged)	−0.136*** (0.014)	−0.106*** (0.013)	−0.106*** (0.012)
Population growth	0.127 (0.078)	0.116 (0.077)	0.027 (0.076)
Median household income	−0.001*** (0.000)	−0.001** (0.000)	−0.001** (0.000)
Log permits (lagged)	0.006*** (0.002)	0.006*** (0.002)	0.004*** (0.002)
<i>Composition controls (vary by specification):</i>			
Avg. borrower LTI (lagged)	0.088*** (0.008)	0.072*** (0.007)	
Avg. borrower LTI (contemporaneous)			0.131*** (0.007)
Conforming share (lagged)		0.083*** (0.010)	
Conforming share (contemporaneous)			0.126*** (0.010)
FHA share (lagged)		0.164*** (0.021)	
FHA share (contemporaneous)			0.424*** (0.021)
Minority share (lagged)		0.403*** (0.129)	
Minority share (contemporaneous)	0.422*** (0.131)		0.316*** (0.119)
<i>Panel B: SD-standardized coefficient</i>			
Regulatory Exposure Index $\times$ Post (std.)	0.0122*** (0.0017)	0.0149*** (0.0017)	0.0151*** (0.0016)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	23,130	23,130	23,130
Clusters	2,378	2,378	2,378
Within R-squared	0.046	0.063	0.143
F-statistic	48.95	75.12	86.69

Notes: Estimates from equation (5.1) with county and year fixed effects; dependent variable is non-bank mortgage market share. Panel A reports raw coefficients; Panel B standardises the instrument to one standard deviation (SD = 0.169). Diagnostic statistics are identical across panels. Standard errors clustered at the county level. F-statistic tests the excluded instrument, exceeding the Stock–Yogo (2005) critical value of 16.38 throughout. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Counties: 2,378.

6.2 Second Stage: Non-Bank Share and Local House Price Dynamics

Table 5 reports the 2SLS estimates of the effect of non-bank mortgage market share on three housing market outcomes: three-year house price volatility, annual house price growth, and three-year cumulative house price growth. All specifications include the baseline controls described in Appendix C together with county and year fixed effects. The estimated effects should be interpreted as the impact of non-bank mortgage expansion on housing market outcomes for counties whose non-bank mortgage market share increased because they were exposed to weakly capitalised banks prior to Basel III.

Table 5. Housing Market Outcomes: Volatility and Growth

	(1) HPV (3yr)	(2) HP Growth	(3) 3yr Cum. Growth
Non-bank mortgage share	−0.055* (0.028)	0.791*** (0.121)	2.021*** (0.303)
1 SD effect (pp)	−0.874	12.658	32.356
Sample mean	0.0287	0.0084	0.0138
Observations	23,130	23,130	23,130
F-statistic	48.96	48.96	48.96

Notes: Estimates from equation (5.2), instrumented by the REI interacted with Post. All specifications include county and year fixed effects and the full control set described in Appendix C: lagged average borrower LTI, unemployment rate, lagged log HPI, population growth, median household income, minority share, and lagged log permits. Standard errors, clustered at the county level, are in parentheses. F-statistic is the Kleibergen–Paap rk Wald F-statistic on the excluded instrument. 1 SD effect is the coefficient multiplied by the sample standard deviation of non-bank share (0.1601), expressed in percentage points. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Column (1) shows that higher non-bank mortgage market share reduces house price volatility. A one percentage point increase in non-bank mortgage share reduces the three-year rolling standard deviation of house price growth by 0.055 percentage points ($p < 0.10$). In economic terms, a one standard deviation increase in non-bank mortgage share reduces house price volatility by approximately 0.9 percentage points. Since the average three-year house price volatility is 2.9%, this corresponds to a reduction of about 30% relative to the sample average.

Column (2) shows that a one percentage point increase in non-bank mortgage share raises annual house price growth by 0.791 percentage points ($p < 0.01$). In standardized terms, a one standard deviation increase in non-bank mortgage share raises annual house price growth by approximately 12.7 percentage points. This corresponds to about 2.5 standard deviations of annual house price growth, indicating a substantial increase in house price appreciation in counties with greater non-bank expansion.²¹

The estimates in the last column show that a one percentage point increase in regulation-induced

²¹The sample mean annual growth rate is close to zero (0.8%), hence the effect is not expressed as a percentage of the mean.

non-bank mortgage share raises three-year cumulative house price growth by 2.021 percentage points ($p < 0.01$). In standardized terms, a one standard deviation increase in non-bank mortgage share raises three-year cumulative growth by approximately 32.4 percentage points. This corresponds to about 2.8 standard deviations of three-year cumulative growth, suggesting a substantial cumulative effect in counties with greater non-bank expansion.²²

Additional housing stability outcomes, Peak-to-trough Drawdown and Price growth persistence are reported in Appendix Table 20.

6.3 Borrower Leverage

Table 6 reports IV estimates of the effect of non-bank mortgage share on the county average loan-to-income (LTI) ratio. Since borrower leverage may differ across lender types and loan product types, the analysis also estimates the effect separately for these categories.

Non-bank mortgage expansion increases borrower leverage. Column (1) shows that a one percentage point increase in non-bank mortgage share raises the county average LTI ratio by 1.366 ($p < 0.01$). In economic terms, a one standard deviation increase in non-bank mortgage share raises the average LTI ratio by 0.219, equivalent to a 9.8% increase relative to the sample mean of 2.243.

Table 6. Average Loan-to-Income Ratio: By Lender Type

	(1) Overall	(2) Non-bank	(3) Bank
Non-bank mortgage share	1.366*** (0.215)	1.425*** (0.300)	1.080*** (0.244)
Observations	23,130	23,129	23,130
F-statistic	44.94	44.96	44.94
% change (1 SD)	9.8	9.3	8.2

Notes: Estimates from equation (5.2), instrumented by the REI interacted with Post. Dependent variable is average loan-to-income ratio, defined overall and by lender type. All specifications include county and year fixed effects and the control set described in Appendix C, excluding lagged average LTI to avoid circularity with the outcome variable. Standard errors, clustered at the county level, are in parentheses. F-statistic is the Kleibergen–Paap rk Wald F-statistic on the excluded instrument. Percent change (1 SD) is the percentage change in the outcome from a one standard deviation increase in non-bank share, relative to each column’s own sample mean; sample size varies slightly across columns due to outcome-specific missingness. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The coefficients in columns (2) and (3) show how this effect differs for non-bank and bank loans. Both coefficients are positive and significant at the 1% level. A one percentage point increase in non-bank mortgage share raises the average LTI ratio of non-bank loans by 1.425 and that of bank loans by 1.080. In economic terms, a one standard deviation increase in non-bank mortgage share raises average LTI by 9.3% relative to the mean for non-bank loans, compared with 8.2% for bank loans. The increase in leverage is also evident across major loan product types. Table 7

²²The sample mean three-year cumulative growth rate is close to zero (1.4%), so the effect is not expressed as a percentage of the mean.

shows that a one standard deviation increase in non-bank mortgage share raises average LTI by 9.8% for conforming loans, 12.1% for non-conforming loans, and 13.2% for FHA loans, with all estimates significant at the 1% level. The largest effect is observed for FHA lending. The mechanisms behind the stronger effect in FHA lending are examined in Chapter 7.

Table 7. Average Loan-to-Income Ratio: By Product Type

	(1) Overall	(2) Conforming	(3) Non-conf.	(4) FHA
Non-bank mortgage share	1.366*** (0.215)	1.332*** (0.250)	1.512*** (0.288)	2.111*** (0.390)
Observations	23,130	23,130	23,129	23,124
F-statistic	44.94	44.94	45.02	44.90
% change (1 SD)	9.8	9.8	12.1	13.2

Notes: Estimates from equation (5.2), instrumented by the REI interacted with Post. Dependent variable is average loan-to-income ratio, defined overall and by loan product type. All specifications include county and year fixed effects and the control set described in Appendix C, excluding lagged average LTI to avoid circularity with the outcome variable. Standard errors, clustered at the county level, are in parentheses. F-statistic is the KP F-statistic on the excluded instrument. Percent change (1 SD) is the percentage change in the outcome from a one standard deviation increase in non-bank share, relative to each column's own sample mean; sample size varies slightly across columns due to outcome-specific missingness. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8. Share of Loans Exceeding the 4.5x Loan-to-Income Threshold

	(1) High-LTI Share
Non-bank mortgage share	0.249*** (0.0447)
Observations	23,130
F-statistic	44.94
% change (1 SD)	79.8

Notes: Estimates from equation (5.2), instrumented by the REI interacted with Post. The specification includes county and year FE and the control set described in Appendix C. Standard errors, clustered at the county level, are in parentheses. F-statistic is the KP F-statistic on the excluded instrument. % change (1 SD) is the percentage change in the outcome from a one standard deviation increase in non-bank share, relative to the sample mean. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Non-bank expansion also raised the share of loans exceeding a high-leverage threshold. Table 8 reports the effect on the share of loans with an LTI ratio at or above 4.5.²³ The coefficient on non-bank share is 0.249, significant at the 1% level. A one standard deviation increase in non-bank share raises the high-LTI share by approximately 4.0 percentage points, equivalently an 79.8% increase relative to the sample mean. This is the largest proportional effect of any outcome examined in this thesis. It suggests that non-bank expansion did not simply increase average borrower leverage, it also increased the prevalence of highly leveraged mortgages.

²³Following the threshold used by the Bank of England's Financial Policy Committee (2014) and, similarly, the Bank of Canada (Bilyk et al., 2020), for identifying highly leveraged mortgages.

6.4 Credit Risk Pricing

Table 9 reports IV estimates of non-bank share on mortgage credit rate spreads. The analysis examines the overall county-level average spread, spreads by lender type, the share of higher-priced loans, and purchase-loan spreads. All specifications use the same controls as Section 6.3.

Non-bank mortgage expansion increases credit rate spreads. Column (1) shows, a one standard deviation increase in non-bank share raises the average rate spread by 18.1% relative to the sample mean ($p < 0.01$). Columns (2) and (3) show the effect differs across lender types. The increase in rate spreads is larger for non-bank loans, rising by 45.8% relative to the mean ($p < 0.01$), compared with 16.2% for bank loans ($p < 0.01$). This suggests that the pricing of non-bank loans also becomes relatively more costly in counties experiencing greater non-bank growth.

Table 9. Credit Rate Spread and Higher-Priced Loan Share

	(1) Total	(2) Non-bank	(3) Bank	(4) HP share	(5) Purchased
Non-bank mortgage share	2.678*** (0.644)	6.705*** (1.339)	2.452*** (0.644)	0.222*** (0.055)	0.978** (0.446)
Observations	23,047	22,254	22,702	23,130	22,486
F-statistic	44.38	44.96	43.57	44.94	44.70
% change (1 SD)	18.1	45.8	16.2	62.3	7.3

Notes: Estimates from equation (5.2), instrumented by the REI interacted with Post. Columns (1)–(3) report the volume-weighted average credit rate spread among loans classified as higher-priced, overall and by lender type. Column (4) reports the volume-weighted share of all lending classified as higher-priced. Column (5) reports the average rate spread for purchase loans specifically. All specifications include county and year fixed effects and the control set used for credit outcomes. Standard errors, clustered at the county level, are in parentheses. % change (1 SD) is the percentage change in the outcome from a one standard deviation increase in non-bank share, relative to each column’s own sample mean; sample size varies across columns due to outcome-specific missingness. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Column (4), the higher-priced share shows an even stronger increase. A one standard deviation increase in non-bank share raises the higher-priced loan share by 3.55 percentage points, equivalent to a 62.3% increase relative to the sample mean. This implies non-bank expansion raises both how many loans carry elevated rates and how elevated those rates are.

Finally, the pattern extends to purchase loans, where a one standard deviation increase in non-bank share raises the purchased loan rate spread by 7.3% relative to the sample mean ($p < 0.01$). In sum, these results suggest that counties experiencing greater non-bank mortgage expansion also experienced wider credit rate spreads and a larger share of higher-priced lending.

6.5 Affordability Pressure

Table 10 reports IV estimates of the effect of non-bank share on the log price-to-income ratio. This measure proxies housing affordability by comparing house prices with local household income. The coefficient on non-bank share is 0.677, significant at the 1% level. In economic terms,

a one standard deviation increase in non-bank mortgage share raises the log price-to-income ratio by 0.108, corresponding to an increase of approximately 11.4% in counties with high exposure to weak banks. This deterioration reflects house price growth outpacing income growth rather than income falling in absolute terms.

Table 10. Effect of Non-Bank Share on Housing Affordability

	(1) Log Price-to-Income Ratio
Non-bank mortgage share	0.677*** (0.116)
Observations	23,130
F-statistic	42.89
% change (1 SD)	11.4

Notes: Estimates from equation (5.2), instrumented by the REI with Post (Section 5.3). Dependent variable is the log price-to-income ratio, $\log(\text{house price index})$ minus $\log(\text{median household income})$. Median household income is excluded from the control set since it enters the outcome's construction directly. All specifications include county and year FE. Standard errors, clustered at the county level, are in parentheses. F-statistic is the KP F-statistic on the excluded instrument. % change (1 SD) is computed as $100(e^{\hat{\beta} \times \text{SD}} - 1)$, the appropriate transformation for a log-linear outcome, reflecting the percentage change in the price-to-income ratio from a one standard deviation increase in non-bank share. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In sum, Sections 6.2–6.5 reveal a story in which non-bank expansion, driven by regulatory pressure on weak banks, reduced medium-run house price volatility, and increased house price growth, while also being associated with higher borrower leverage, wider credit rate spreads, and a higher price-to-income ratio. These findings suggest that the expansion of non-bank lending may have supported local housing markets in the medium run while simultaneously building up leverage, rate spreads and affordability pressure that could pose risks later, discussed in detail in Chapter 9.

6.6 Heterogeneity by Minority Share

This section examines whether the effects of Basel III varied with counties' baseline minority share. In Table 11, the coefficient on interaction of REI and Post is 0.052***. Precisely, this means a one-standard-deviation increase in regulatory exposure raises the non-bank share by 0.88 percentage points in a county with a zero baseline minority population share. This effect grows larger in counties with a higher baseline minority population share. It increases by a further 0.24 percentage points for every 10 percentage point increase in baseline minority share.²⁴

Table 11 suggests higher baseline minority share strengthens the volatility-reducing effect of non-bank mortgage expansion. A one standard deviation increase in non-bank mortgage share reduces house price volatility by 0.84 percentage points in a county with zero baseline minority population share, an effect that grows by a further 0.16 percentage points for every 10 percentage point increase in baseline minority share.²⁵ The same increase raises annual house price growth

²⁴Calculated as $\pi_1 \times \text{SD}(\text{exposure})$ and $\pi_2 \times \text{SD}(\text{exposure})$, with $\pi_1 = 0.0518$, $\pi_2 = 0.140$, and $\text{SD}(\text{exposure}) = 0.1691$.

²⁵Calculated as $\beta_1 \times \text{SD}(\text{nonbank_share})$ and $\beta_2 \times \text{SD}(\text{nonbank_share})$, with $\beta_1 = -0.052$, $\beta_2 = -0.100$,

by 12.65 percentage points on its own, rising by a further 0.23 percentage points for every 10 percentage point increase in baseline minority share. It further raises three-year cumulative house price growth by 31.72 percentage points in a county with zero baseline minority share, rising by a further 0.73 percentage points for every 10 percentage point increase in minority share.

Lastly, a one standard deviation increase in non-bank mortgage share raises the log price-to-income ratio by 0.107 in a county with zero baseline minority population share, rising by a further 0.005 for every 10 percentage point increase in baseline minority share.

Table 11. Heterogeneity by Baseline Minority Share: First Stage, Housing Outcomes, and Affordability

	(1) First Stage	(2) HPV (3yr)	(3) HP Growth	(4) 3yr Cum. Growth	(5) Log Price-to-Income
REI $\times$ Post	0.052*** (0.011)				
REI $\times$ Post $\times$ Baseline Minority Share	0.140*** (0.027)				
Non-bank share		-0.052* (0.029)	0.790*** (0.126)	1.981*** (0.312)	0.665*** (0.123)
Non-bank share $\times$ Baseline Minority Share		-0.100*** (0.020)	0.142* (0.074)	0.457** (0.180)	0.337*** (0.076)
Observations	23,130	23,130	23,130	23,130	23,130
F-statistic	20.51 / 27.27	27.12	27.12	27.12	24.97

Notes: The two F-statistics reported are for REI $\times$ Post and its interaction with baseline minority share, respectively. All specifications include county and year FE and the control set described in Appendix C, excluding minority share as a standalone control since it enters the interaction term. Standard errors, clustered at the county level, are in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In sum, the results together suggest that the housing market and price-to-income effects of non-bank mortgage expansion are stronger in counties with larger baseline minority populations. In other words, the effects of Basel III-induced non-bank mortgage expansion were not uniform across local housing markets, but were considerably more pronounced in counties with larger pre-existing minority populations.

7 Mechanisms

7.1 The Reallocation: Non-Bank Expansion

Having established the effects of Basel III-induced non-bank mortgage expansion, this chapter examines the mechanisms underlying these findings. It first investigates why non-bank mortgage share increased following Basel III in counties differentially exposed to weak banks, and the channels through which this expansion influenced housing market outcomes, borrower risk, and affordability. It concludes by exploring why these effects were more pronounced in counties with larger baseline minority populations.

In the first stage, the logic is that counties highly exposed to weak banks would see more

$$SD(\text{nonbank_share}) = 0.1601.$$

non-bank share growth than less-exposed or unexposed counties. This is because weak banks that are capital constrained, or that have thin capital buffers, have to adjust their T1RBC ratio to meet the minimum requirements set by Basel III, and this leads to a shrinking of mortgage loans. They cut mortgage loans because mortgages carry higher risk weights, which pulls down the T1RBC ratio. If this mechanism holds, three things should be true: weak banks' T1RBC ratios should rise, their mortgage lending should fall, and non-bank lending should rise in their place. The reduced-form regressions (Appendix Table 23) show that weak banks raised T1RBC ratios more than other banks following Basel III's implementation, bank lending fell in high weak bank exposure counties post-2013, and non-bank lending rose over the same period.

In addition, Appendix Tables 21 and 22 examine bank and non-bank lending by loan type. Bank lending declined in both conventional and FHA loan categories, but the contraction was substantially larger for FHA (-53.2%) and conforming loans (-22.2%).²⁶

This is consistent with a second channel through which non-banks expanded: Mortgage Servicing Rights (MSRs). MSRs generated when conforming and FHA loans are sold with servicing retained are subject to deduction from capital above a defined threshold. Banks may only count MSRs up to 10% of Common Equity Tier 1 (CET1) capital, with any excess deducted directly from CET1, creating capital pressure specifically tied to the loan types banks predominantly originate for sale, which in turn affects the T1RBC ratio. Although this thesis does not examine this mechanism, Buchak et al. (2018) document the effect of this specific channel on non-bank share growth.

7.2 From Reallocation to Local Housing Market Outcomes

Table 5 reports IV estimates of non-bank share on three house price outcomes: three-year rolling volatility (HPV), annual house price growth, and three-year cumulative growth.

As banks reduced mortgage lending following Basel III, non-bank lenders more than offset this contraction, leading to a net increase in mortgage credit in high-exposure counties (Table 3). One plausible explanation is that the resulting expansion in credit supply may have relaxed borrowing constraints for households seeking to purchase homes, allowing pre-existing unmet housing demand to translate into effective demand and placing upward pressure on house prices. This mechanism is explained by the broader credit supply literature, which finds that exogenous expansions in mortgage credit translate into higher house prices (Mian and Sufi, 2009; Favara and Imbs, 2015; Choi et al., 2024). Housing supply also responds to this credit expansion: building permits, a measure of new construction activity, rise significantly with non-bank share. This indicates that the price effect reported here does not reflect a demand effect estimated under an assumption of fixed supply.

The three-year cumulative growth specification serves primarily as a persistence check. The stronger effect on three-year cumulative house price growth reflects the persistence of the underlying credit reallocation. As shown in Section 6.1, non-bank lenders did not merely replace bank lending in the initial phase of Basel III; the increase in non-bank mortgage activity persisted

²⁶Conforming loans are a subset of conventional loans; conventional loans include both conforming and non-conforming loans.

throughout the post-2013 period. This continued availability of housing credit could have allowed house prices to appreciate further, rather than reversing after an initial increase. In other words, the effect of Basel III-induced non-bank mortgage expansion on house prices was persistent rather than transitory.

The reduction in medium-run house price volatility can be explained by a credit-supply-smoothing mechanism. Basel III reduced bank mortgage lending in high-exposure counties, creating the potential for an abrupt contraction in housing credit. However, the expansion of non-banks in these counties appears to have reduced this disruption by maintaining, and ultimately increasing, the availability of mortgage credit. Rather than allowing house prices to adjust sharply to a sudden withdrawal of credit, non-bank lending may have smoothed the local credit supply, resulting in more gradual medium-run house price movements and lower medium-run volatility. However, this thesis does not test this smoothing channel directly.

In sum, the evidence suggests that Basel III-induced non-bank share expansion appears to have stabilised local housing markets in the medium run by smoothing the contraction in bank mortgage lending while simultaneously supporting sustained house price appreciation.

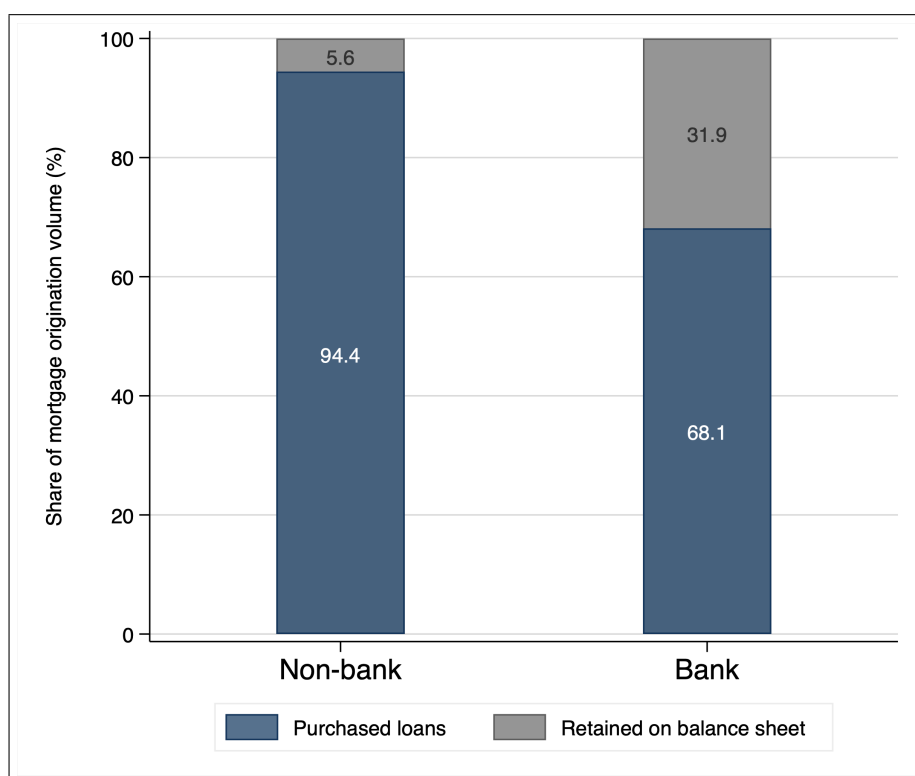
7.3 From Reallocation to Borrower Leverage

Table 6 and Table 8 suggest that the reallocation of mortgage lending following Basel III affected not only the average level of borrower leverage but also the distribution of leverage within local mortgage markets.

One explanation relates to the business model of non-bank lenders. Unlike banks, non-banks originate mortgages primarily for sale into secondary markets rather than retaining them on their balance sheets. Figure 4 shows that around 94% of non-bank loans are purchased loans, and Figure 5 further shows that these loans are predominantly sold through secondary-market channels, whereas banks retain a substantially larger share of their originations. Non-bank loans sold in this way inherently carry a higher baseline LTI ratio. This mechanism is further supported by Keys et al. (2010), who show that mortgages originated for sale are subject to weaker screening incentives than loans retained on lenders' balance sheets. Therefore, the expansion of non-bank lending is consistent with the observed increase in both average LTI and the share of highly leveraged mortgages.

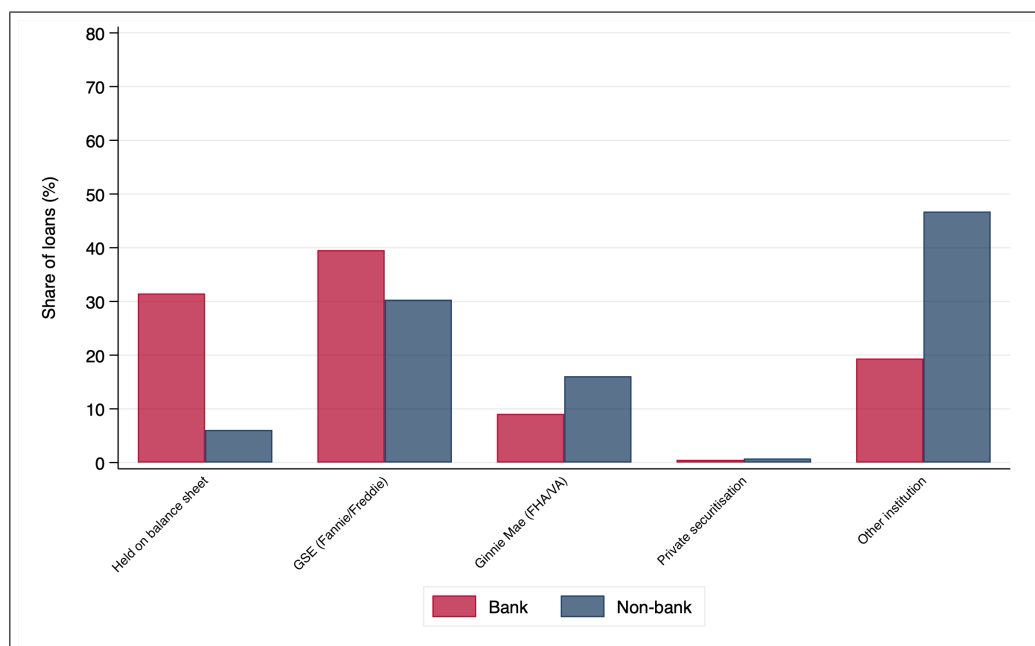
However, this channel alone cannot explain the results. Although the increase in average LTI is larger for non-bank lenders, average LTI ratio also rises among banks. If the effect reflected only the growth of non-bank lending, average bank LTI would remain broadly unchanged. Instead, the evidence suggests that borrower risk was reallocated across lenders. Lim et al. (2024) document a similar mechanism in a different regulatory setting, showing that when a regulatory change drives some lenders out, the remaining lenders who do not face the same regulatory pressure compete for the displaced borrowers, reallocating risk rather than reducing it overall. This thesis does not test this displacement mechanism directly. However, the observed retreat of banks from FHA and conforming lending, where borrower leverage is typically higher, together with rising LTI among both non-bank lenders and remaining banks, is consistent with, though not

Figure 4. Share of Bank and Non-Bank Originations Sold as Purchased Loans



Notes: Volume-weighted share of mortgage origination sold to secondary market purchasers (purchased loans) versus retained on the originator's balance sheet, by lender type. Purchased loans include those sold to GSEs (Fannie Mae, Freddie Mac, Ginnie Mae) and private investors. Bank is a depository institution; non-bank is a mortgage company or other non-depository lender (Section 4.2.1). Source: HMDA LAR, 2009–2018.

Figure 5. Distribution of Purchaser Type Among Sold Loans



Notes: Share of mortgage originations by purchaser type and lender category. Bank is a depository institution (other_lender_code = 0); non-bank is a mortgage company or other non-depository lender (Section 4.2.1). GSE is a government-sponsored enterprise (Fannie Mae or Freddie Mac). Ginnie Mae purchases FHA and VA insured loans. Source: HMDA LAR, 2009–2018.

direct evidence for, these displaced borrowers being absorbed elsewhere in the market (Table 6, Appendix tables 21 and 22).

The increase in leverage is particularly pronounced for FHA mortgages. Banks experienced their largest lending contraction in the FHA segment following Basel III, creating the greatest scope for borrower reallocation. At the same time, FHA mortgages are specifically designed to serve borrowers with lower down payments and higher leverage. The interaction of these two forces produced the largest increase in average LTI among the three loan categories (Table 7). By contrast, conforming mortgages experienced the smallest increase in average LTI, while borrower reallocation also occurred in this segment, conforming mortgages remained subject to the underwriting and eligibility standards required for GSE purchase, so leverage increased by less than in the FHA and non-conforming segments.

7.4 From Reallocation to Credit Risk Pricing

Table 9 reports the estimates of non-bank share on credit rate spreads and the higher-priced loan share. The increase in credit rate spreads is consistent with the rise in borrower leverage discussed in Section 7.3. Since mortgage interest rates incorporate borrower risk, lenders would be expected to charge wider spreads to a more leveraged borrower pool. The increase in mortgage spreads for both bank and non-bank loans is therefore consistent with a market-wide, risk-based pricing response to higher borrower leverage.

A further explanation, though not tested here, is that non-banks' funding source also contributes to the larger increase in their rate spreads. Shoemaker (2019) notes that, unlike banks, non-bank lenders do not fund mortgages through insured deposits. Instead, they rely primarily on warehouse credit and mortgage-backed securitisation, a funding source that can become more expensive and less accessible during periods of financial market stress. Through the same mechanism, as more mortgages are originated to highly leveraged borrowers and priced at wider spreads, a larger fraction exceed HMDA's threshold for classification as higher-priced loans. Though this mechanism is plausible, it is not directly tested.

The larger increase in spreads among purchased loans is also consistent with the originate-to-distribute business model. As discussed in Section 7.3, this model has been associated with weaker screening incentives (Keys et al., 2010). Hence, the expansion of purchased lending following Basel III likely involved relatively riskier borrowers, resulting in a larger increase in rate spreads.

7.5 From Reallocation to Affordability Pressure

Table 10 reports IV estimates of the effect of non-bank share on the log price-to-income ratio. The building pressure on housing affordability is a combined outcome of the mechanisms discussed in the previous sections. Basel III-induced non-bank expansion led to a rise in house price growth. However, the results in Appendix Table 24 suggest that the increase in house prices was substantially larger than the increase in income. A 1SD increase in non-bank share raises the level of house prices by 15.4%. Over the same increase, median household income rises by

3.9%. The price effect is roughly four times the income effect. The resulting rise in the log price-to-income ratio therefore reflects the combined effect of stronger house price appreciation together with comparatively modest income growth.

7.6 From Reallocation to Heterogeneity by Minority Share

Table 11 suggests that the first-stage relationship is stronger in counties with higher baseline minority share. To study the mechanism, I split counties above and below the median baseline minority share. The descriptive statistics in Appendix Table 25 suggest that in 2008 the Regulatory Exposure Index was nearly identical across the two groups: 0.275 in high-minority counties, compared with 0.278 in low-minority counties. High-minority counties did not, therefore, have greater weak-bank exposure to begin with. What did differ was the pre-existing lender composition. Table 25 also shows that in 2008, non-bank mortgage market share was already higher in high-minority counties (34.8%) than in low-minority counties (28.7%), and bank market share was lower in high-minority counties (65.2%) than in low-minority counties (71.3%). Given the same weak-bank exposure, this suggests high-minority counties already had a larger established non-bank presence and a smaller established bank presence, positioning them to expand further once weak banks retreated, while low-minority counties, with a smaller pre-existing non-bank foothold and a deeper bank presence, had less capacity to respond to the same shock as quickly or as extensively.

Another explanation for the larger non-bank expansion in high-minority counties is that these counties already had a higher share of FHA lending (Table 25: 21.7%, compared with 17.6% in lower-minority counties). As discussed in Section 7.3, Basel III led to the sharpest contraction in weak-bank FHA lending, and non-banks expanded largely through FHA lending. This implies that high-minority counties, which relied more heavily on FHA lending before Basel III, experienced a larger increase in non-bank mortgage share after the regulation.

Section 6.6 showed that the housing market effects of non-bank mortgage expansion are stronger in counties with larger baseline minority populations. This could simply reflect the larger first-stage increase in non-bank share in these counties, driven by pre-existing differences in lender composition and FHA shares. However, the interaction estimates show the marginal effect also differs: for the same increase in non-bank share, house price growth rises more, volatility falls more, and the price-to-income ratio increases more in counties with higher baseline minority share.

One interpretation, though not directly tested, is that counties with larger baseline minority populations may differ in underlying housing or mortgage market characteristics that make house prices more responsive to additional mortgage credit. The results should therefore be interpreted as documenting systematic heterogeneity in the effects of non-bank mortgage expansion, rather than as identifying a single underlying mechanism. The next chapter examines the robustness of these results.

8 Robustness Checks

8.1 Alternative Instrument Specifications

To show that the effect does not depend on how the Regulatory Exposure Index (REI) itself is built, I estimate two alternative instrument construction specifications, together with an alternative weak-bank threshold.

The first alternative sets the weak-bank threshold to 8.5%, matching Basel III’s minimum T1RBC ratio plus its capital conservation buffer.²⁷ Column (2) of Table R1 shows the first-stage results with this threshold. In standardized terms, a one standard deviation increase in exposure under this alternative threshold raises non-bank share by 1.18 percentage points, close in magnitude to the 1.22 percentage points estimated under the main 8% threshold, with an F-statistic of 47.44, nearly matching the main specification’s 48.95. Across all three constructions, the coefficient is positive and significant at the 1% level, confirming that the effect is not driven by the specific 8% threshold used in the main specification.

Second, I use a continuous measure of exposure to weak banks: the instrument is the county-level weighted average of the inverse T1RBC ratio across banks, interacted with the post-2013 indicator. Column (3) of Table R1 shows the first-stage results with this specification. In standardized terms, a one standard deviation increase in this continuous exposure measure raises non-bank share by 0.81 percentage points, compared to 1.22 percentage points under the main binary specification. The continuous measure produces a smaller but directionally consistent effect, with an F-statistic of 24.75.

The two binary-threshold specifications (8% and 8.5%) remain close in magnitude, and the continuous measure is directionally consistent but somewhat weaker. This confirms that the effect is not driven by the specific threshold or functional form used to classify weak banks in the main specification.

8.2 Sensitivity to Treatment Timing

As a further check, I redefine the post-treatment period using 2012 and 2014 as alternative cutoff years instead of 2013. Table R2 reports these estimates alongside the main specification. The estimates are very similar across the three definitions. A one standard deviation increase in exposure raises non-bank share by 1.19 percentage points using the 2012 cutoff, 1.22 percentage points using the 2013 cutoff, and 1.17 percentage points using the 2014 cutoff. The corresponding first-stage F-statistics are 45.27, 48.95, and 45.28. All coefficients in the first stage are positive and significant at the 1% level.

The stability of the estimates across nearby definitions of the post-treatment period suggests that the first-stage result is not driven by the precise placement of the post-period boundary.

²⁷See Section 5.2.2 for discussion of why the 8% threshold, rather than the literal 8.5% regulatory minimum, is used in the main specification.

Table R1. Sensitivity to Instrument Threshold and Functional Form

	(1) Main (8%)	(2) 8.5% Threshold	(3) Continuous
Regulatory Exposure Index $\times$ Post (std.)	0.0122*** (0.00175)	0.0118*** (0.00171)	0.0081*** (0.00163)
Observations	23,130	23,130	23,130
F-statistic	48.95	47.44	24.75

Notes: Estimates from equation (5.1), with the instrument standardized to 1SD in each specification. All specifications include county and year fixed effects and the control set described in Appendix C. Standard errors, clustered at the county level, are in parentheses. F-statistic tests the excluded instrument, and exceeds the Stock–Yogo (2005) weak-instrument critical value of 16.38 throughout. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table R2. Robustness: Alternative Post-Period Cutoffs

	(1) Post $\geq$ 2012	(2) Post $\geq$ 2013 (main)	(3) Post $\geq$ 2014
Regulatory Exposure Index $\times$ Post (std.)	0.0119*** (0.0018)	0.0122*** (0.0017)	0.0117*** (0.0017)
Observations	23,130	23,130	23,130
F-statistic (excl. instrument)	45.27	48.95	45.28

Notes: Each column replicates the main first-stage specification (Table 4, Panel B, Column 1) with a different post-period cutoff year instead of 2013, with the instrument standardized to one standard deviation. All specifications include county and year FE and the control set described in Appendix C. Standard errors, clustered at the county level, are in parentheses. F-statistic tests the excluded instrument via `test`, not the joint model F. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

This is consistent with the event-study results in Section 5.4.1, which show that non-bank share remains elevated over several years rather than reverting after a single year. The event study and placebo test in Sections 5.4.1 and 5.4.2 remain the appropriate evidence for establishing that the effect specifically originates in 2013.

8.3 Sensitivity to Confounders

State $\times$ year fixed effects for the first stage. As a further check, I combine county fixed effects with state $\times$ year fixed effects instead of year fixed effects alone. Since the REI depends on the location of weak banks in 2008, and bank presence is not randomly distributed geographically, state $\times$ year fixed effects test whether the first-stage relationship remains significant after accounting for state-specific shocks in each year. The specification drops 16 singleton observations, so total observations fall to 23,120 across 2,377 counties. Table R3 shows that a one standard deviation increase in exposure raises non-bank share by 0.88 percentage points, compared with 1.22 percentage points in the main specification. Although the coefficient is smaller, the relationship remains positive and significant at the 1% level. This specification is more demanding than the baseline first stage, because state $\times$ year fixed effects absorb variation across states within each year, leaving only within-state variation in exposure to identify the coefficient. The result therefore continues to support the regulatory channel after accounting for state-specific changes over time.

Additional Sensitivity-to-Confounders robustness checks are reported in Appendix F.

Table R3. Robustness: State $\times$ Year Fixed Effects

	(1) State $\times$ Year FE
Regulatory Exposure Index $\times$ Post (std.)	0.0088*** (0.0020)
Observations	23,120
F-statistic	19.42

Notes: County fixed effects combined with state-by-year fixed effects instead of year fixed effects alone, testing whether the first-stage relationship remains significant after accounting for state-specific shocks in each year. The instrument is standardized to one standard deviation. All specifications include the control set described in Appendix C. Standard errors, clustered at the county level, are in parentheses. F-statistic tests the excluded instrument, above the Stock–Yogo (2005) weak-instrument threshold of 16.38. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9 Discussion

Basel III-induced non-bank mortgage expansion is associated with more stable local house price dynamics. The reduction in medium-run house price volatility and the stronger, more consistent house price growth can be viewed as favourable outcomes when considered in isolation. Lower volatility implies more predictable housing values, reducing uncertainty for households and lenders. Conversely, sharp and unpredictable price declines can reduce collateral values and increase financial pressure, especially for leveraged borrowers. Overall, the evidence points to a

stabilising effect on local housing markets.

The implications become more nuanced when considering how this credit was supplied. As banks reduced mortgage lending following Basel III, non-bank lenders expanded, maintaining and ultimately increasing the supply of mortgage credit. This reallocation helps explain how housing-market activity could remain strong despite the contraction in bank lending. But the effect of the reallocation extends beyond the quantity of credit supplied: who originates the mortgage matters for the conditions under which credit is provided. The shift toward non-bank lending is associated with higher borrower leverage, a larger share of high-LTI lending, and wider credit rate spreads. Thus, the mortgage market did not simply replace lost bank credit with an equivalent supply of non-bank credit; the composition and pricing of mortgage credit changed alongside its expansion.

The increase in leverage is particularly relevant because leverage determines how exposed household balance sheets are to changes in housing wealth. Mian et al. (2013) show that highly leveraged households experience larger balance-sheet effects from housing-wealth shocks and reduce consumption more sharply in response. In this thesis, the rise in leverage therefore represents a less favourable credit-market condition alongside the observed housing-market stabilisation. Importantly, this does not imply that the additional leverage resulted in financial distress; rather, it identifies a change in household balance-sheet exposure that may matter when housing conditions deteriorate.

The increase in credit rate spreads provides a further indication that the expansion of mortgage credit was accompanied by changes in credit conditions. Part of the rise, occurring alongside rising leverage, reflects a rise in borrower risk. In addition to this, the rise in the log price-to-income ratio, a simple affordability proxy used, cuts both ways: it implies existing homeowners become better off, since rising house prices increase the value of their housing assets, but it makes homeownership more difficult for households seeking to enter the market, since at a given level of income a prospective buyer faces a higher house price relative to what they earn.

The heterogeneity results make the interpretation more consequential. These results should not be interpreted as evidence that minority households themselves experienced different treatment, since the analysis is conducted at the county level. Rather, they show that the consequences of a common national regulatory change depended on pre-existing local characteristics. The institutional structure of local mortgage markets therefore shaped how strongly the benefits and costs of the regulatory reallocation were experienced.

Taken together, these results form the basis for the thesis's interpretation of stabilisation with latent fragility. Stabilisation in housing-market conditions, alongside latent fragility in credit conditions reflected in higher leverage, wider spreads, and greater affordability pressure. The results only show that conditions associated with increased sensitivity to a future shock are present, and concentrated in specific counties and specific segments of the leverage distribution. Therefore, the distinction between risk that is present and harm that may actually materialise is important for interpreting these results.

This suggests one broader policy consideration: capital regulation confined to depository in-

stitutions may relocate mortgage-market risk-taking rather than reduce it, when a substantial share of lending can shift to less-regulated entities. In this setting, the strengthening of bank balance sheets under Basel III was accompanied by a reallocation of mortgage lending toward non-banks, together with higher borrower leverage and wider credit rate spreads. This suggests that the effects of capital regulation may depend not only on how strongly banks themselves are regulated, but also on how easily lending can move outside the regulated banking sector.

10 Conclusion

This thesis studies the effect of Basel III capital requirements on non-bank mortgage expansion, and the consequences of that expansion for local housing markets and credit conditions. Using a shift-share instrument based on 2008 county-level exposure to weak banks, I find that non-bank mortgage share rose significantly more in counties with greater exposure following Basel III's 2013 implementation.

This non-bank expansion is associated with lower three-year house price volatility and higher annual and three-year cumulative house price growth. These findings, together with the additional housing stability measures considered in the thesis, indicate that local housing markets in more exposed counties experienced a smoother adjustment following the regulatory change.

At the same time, the expansion is associated with an increase in the average loan-to-income ratio for the mortgage market as a whole, and separately for bank- and non-bank-originated loans. The increase is also present across conforming, non-conforming, and FHA lending, with the largest effect observed for FHA loans, and the share of loans above the high-leverage threshold increases by substantially more than average LTI itself. Credit rate spreads widen for total, non-bank, and bank lending, alongside an increase in the share of lending classified as higher-priced and, importantly, an increase in the rate spread on purchased loans. Finally, housing affordability, proxied by the ratio of house prices to median household income, deteriorates, driven by house prices rising faster than income rather than by income falling.

These effects are also not uniformly distributed. The first-stage relationship between regulatory exposure and non-bank share, and the effects on house price volatility, house price growth, and affordability pressure, are all stronger in counties with a higher baseline minority population share. These differences suggest that the consequences of the regulatory-induced non-bank expansion depend partly on pre-existing local characteristics and the structure of local mortgage markets.

Taken together, I interpret these results as evidence of stabilisation with latent fragility: non-bank expansion cushioned local housing markets against a disruptive credit contraction and supported continued price growth, while simultaneously building leverage, widening credit rate spreads, and increasing pricing pressures. Whether this latent fragility translates into greater macroeconomic vulnerability in the longer run remains an open question, and one beyond the scope of this thesis.

This thesis has three main limitations. Most centrally, I do not observe loan performance or delinquency outcomes, and so cannot test whether the leverage buildup documented here has translated, or would translate, into actual borrower distress; this is why the fragility described

in this thesis is latent rather than demonstrated. I do not examine foreclosure rates, partly because it is unclear when any resulting defaults associated with non-bank expansion would materialise, and partly because of the lack of freely available, matched loan-performance data for non-bank originators. Further, the affordability measure used throughout this thesis is a simple price-to-income ratio that does not incorporate mortgage financing costs directly; a household's actual affordability burden also depends on mortgage interest rates, down payment requirements, household wealth, and individual borrowing constraints. Lastly, the credit conditions studied in this thesis provide evidence only on how the composition of mortgage credit changed in counties experiencing larger increases in non-bank lending, rather than identifying the total effect of Basel III on credit conditions overall.

Considering these limitations, several extensions could be made. The longer-run effects of the latent fragility documented here could be studied using default rates and their broader macroeconomic implications. The effect of non-bank expansion on housing affordability could be examined using the more detailed measures discussed above. Lastly, this thesis could be extended to study broader credit-condition effects at the county level, including denial rates, loan-to-value ratios, and underwriting standards not examined here.

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Appendix

A Additional Institutional Background

Bank and Non-Bank Share

The mortgage market underwent a substantial structural shift in lender composition over the decade after the 2008 financial crisis. Figure 1 shows the average county-level non-bank and bank share of mortgage origination volume from 2009 to 2018.

The figure shows that before 2013, the bank and non-bank shares remained relatively stable, averaging around 31–32% between 2009 and 2012. Beginning in 2013, non-bank share rose sharply and persistently, reaching 42% by 2014 and further increasing to 48–49% by 2017. On the other hand, the exact opposite is happening with bank shares, falling from 68% before 2013 to just over 50% by 2017. This thesis tries to study the effects of this shift caused by a major regulatory change in 2013.

Purchased Loans and Purchaser Types

A loan is described as purchased when it is sold by the originating lender to another entity after origination, rather than retained on the lender’s balance sheet. HMDA records this through the Type of Purchaser field, which reports the type of entity purchasing a covered loan from the lender institution Consumer Financial Protection Bureau, 2015).

Understanding purchaser type is important because whether and to whom a loan is sold has direct implications for underwriting incentives. When a loan is sold to a purchaser with its own underwriting requirements, the originator must meet those requirements for the loan to be eligible for sale. In contrast, loans that are not sold, or are sold to purchasers that impose less stringent screening requirements, may face weaker external underwriting discipline. Keys et al. (2010) show loans destined for sale without a disciplined purchaser show weaker underwriting screening. This distinction is central to the mechanism discussed in the later part of this thesis because non-banks rely disproportionately on selling their originated loans, and a rise in non-bank share has implications for borrower composition at the county level.

HMDA classifies purchasers into nine categories: Fannie Mae, Ginnie Mae, Freddie Mac, Farmer Mac, private securitizer, commercial bank/savings bank/savings association, credit union/mortgage company/finance company, affiliate institution, and other type of purchaser, alongside a code indicating the loan was not sold within the reporting year (Consumer Financial Protection Bureau, 2015).

B Data

Data Sources

The estimation panel draws on the following data sources:

- Home Mortgage Disclosure Act (HMDA) Loan Application Register, Federal Financial Institutions Examination Council ([Federal Financial Institutions Examination Council, 2019b](#)).
- Consolidated Reports of Condition and Income (Call Reports), Schedule RC-R: Regulatory Capital, Federal Financial Institutions Examination Council ([Federal Financial Institutions Examination Council, 2019a](#)).
- House Price Index, county level, Federal Housing Finance Agency ([Federal Housing Finance Agency, 2019](#)).
- Small Area Income and Poverty Estimates (SAIPE), United States Census Bureau ([United States Census Bureau, 2019b](#)).
- County-level unemployment data, United States Department of Agriculture Economic Research Service ([United States Department of Agriculture, Economic Research Service, 2019](#)).
- Annual county population estimates, United States Census Bureau ([United States Census Bureau, 2019a](#)).
- Bridged-race postcensal population estimates, National Cancer Institute SEER Program ([National Cancer Institute, 2019](#)).
- Building Permits Survey, United States Department of Housing and Urban Development ([United States Department of Housing and Urban Development, 2019](#)).
- HMDA Lender Crosswalk (Avery File), [Avery and Bhutta \(2019\)](#).

Bank and Non-bank share. County-year bank and non-bank mortgage shares are calculated as volume-weighted shares of total loan origination:

$$\text{Bank Share}_{ct} = \frac{\sum_i \text{LoanAmount}_{ict} \times \mathbb{I}[\text{other lender code}_{ict} = 0]}{\sum_i \text{LoanAmount}_{ict}} \quad (1)$$

with non-bank share defined analogously as one minus this quantity.

Reasoning for a Three-year window. The choice of a three-year window is based on two considerations. A shorter window would make the measure highly sensitive to individual annual movements, while a substantially longer window could combine housing market conditions that may belong to different phases of the credit cycle. Therefore, the three-year window provides a

Table 15. Summary Statistics: Appendix Variables

	Count	Mean	SD	Min	Max
Non-bank average LTI	23,129	2.466	0.417	0.987	4.735
Bank average LTI	23,130	2.104	0.410	0.835	4.220
High-LTI share ($\geq 4.5x$)	23,130	0.050	0.046	0.000	0.487
Non-bank rate spread	22,254	2.343	0.834	1.560	6.110
Bank rate spread	22,702	2.422	0.708	1.599	5.026
Higher-priced loan share	23,130	0.057	0.045	0.000	0.420
Purchase loan rate spread	23,047	2.372	0.693	1.500	5.330
House price volatility (5yr)	23,130	0.038	0.026	0.007	0.146
Log bank lending volume	23,130	17.988	1.646	13.156	24.427
Log non-bank lending volume	23,130	17.508	1.854	11.513	24.535
Conforming loan share	23,130	0.463	0.131	0.037	1.000
FHA loan share	23,130	0.166	0.075	0.000	0.765

Notes: Summary statistics for variables used only in secondary specifications are reported here. LTI-family variables (rows 1–3) are restricted to the estimation sample of the average LTI IV regression (Table 6, Column 1). Non-bank rate spread, bank rate spread, higher-priced loan share, and purchase loan rate spread (rows 4–7) are each restricted to their own outcome-specific IV regression’s estimation sample. House price volatility (5-year window) is restricted to its own IV regression’s estimation sample. Log bank and non-bank lending volume, conforming loan share, and FHA loan share (rows 9–12) are restricted to the estimation sample of the reduced-form lending volume regression (Table 3). Sample sizes differ slightly across rows because each set of variables is drawn from its own regression’s estimation sample, reflecting outcome-specific missingness.

measure of medium-run housing market volatility that is sufficiently smooth to capture persistent changes in local price dynamics. Because the measure requires three consecutive annual growth observations, the HPI series is extended back to 2005 to ensure that the first observation in the 2009 estimation sample is based on a complete three-year window. As an additional robustness check, I construct an analogous five-year volatility measure using the same procedure.

Annual house price growth. The first outcome measures the annual change in county-level house prices. For county c in year t , annual house price growth is calculated as

$$\text{hpi_log_growth}_{ct} = \log(\text{hpi_2000}_{ct}) - \log(\text{hpi_2000}_{c,t-1}).$$

This transformation measures the approximate percentage change in house prices from one year to the next and provides the underlying growth series used to construct the volatility measure below.

Three-year cumulative house price growth. I build a measure of cumulative house price appreciation over three years (hpi_growth_3yr). This measure captures the overall change in the level of house prices rather than the variability of annual growth:

$$\text{hpi_growth_3yr}_{ct} = \log(\text{hpi_2000}_{ct}) - \log(\text{hpi_2000}_{c,t-3}).$$

The distinction between the two three-year measures is important. hpi_growth_3yr checks whether annual house price growth has been stable or volatile over the preceding three years, whereas hpi_growth_3yr checks how much the level of house prices has increased over the same period. In other words, the former captures stability, while the latter captures cumulative appreciation. Including both allows the empirical analysis to distinguish between a change in the pace of house

price appreciation and a change in the stability of that appreciation.

Additional Housing Stability Measures

Price Growth Persistence. This measure captures the consistency of positive house price growth in a county over recent years, rather than the magnitude or volatility of price changes. For county c in year t , I first define an indicator for whether annual house price growth is positive:

$$Positive\ Growth_{ct} = \mathbb{1}[hpi_log_growth_{ct} > 0] \quad (2)$$

Price growth persistence is then the rolling three-year mean of this indicator:

$$Persistent\ Growth_{ct} = \frac{1}{3} \sum_{s=t-2}^t Positive\ Growth_{cs} \quad (3)$$

This measure ranges from 0 (house prices declined in all three years) to 1 (house prices rose in all three years), and captures the directional consistency of local price movements over the same three-year window used for the main house price volatility measure.

Peak-to-Trough Drawdown. This measure captures the severity of a county's recent house price decline relative to its own recent peak, providing an alternative measure of housing-market downturns. For county c in year t , I first construct the rolling maximum HPI level over a six-year window:

$$Peak\ HPI_{ct} = \max_{s \in [t-5, t]} hpi_2000_{cs} \quad (4)$$

Drawdown is then the proportional decline from this peak to the current level:

$$Draw\ down_{ct} = \frac{Peak\ HPI_{ct} - hpi_2000_{ct}}{Peak\ HPI_{ct}} \quad (5)$$

This measure equals zero when a county is at or above its own six-year peak, and rises toward one as prices fall further below that peak. Unlike house price volatility, which measures dispersion around a local trend, drawdown captures a directional, cumulative decline relative to a recent high.

Finally, I construct a log price-to-income measure as a proxy for housing affordability pressure:

$$\log_price_to_income_{ct} = \log(hpi_2000_{ct}) - \log(MedianHouseholdIncome_{ct}).$$

The measure compares the evolution of local house prices with the evolution of household income. An increase indicates that house prices have risen faster than income, creating greater pressure on housing affordability.

Credit rate spread and high-priced loan share construction

For county c in year t , the volume-weighted average rate spread is calculated as

$$\text{avg_rate_spread}_{ct} = \frac{\sum_i \text{LoanAmount}_{ict} \times \text{RateSpread}_{ict} \times \mathbf{1}[\text{HigherPriced}_{ict}]}{\sum_i \text{LoanAmount}_{ict} \times \mathbf{1}[\text{HigherPriced}_{ict}]},$$

where i indexes individual mortgage loans and $\mathbf{1}[\text{HigherPriced}_{ict}]$ is an indicator equal to one when the loan is classified as higher-priced and zero otherwise.

Higher-priced loan share. This is constructed from the same loan-level classification. I create an indicator for whether each loan is classified as higher-priced. I then calculate the volume of higher-priced lending within each county-year by summing loan_amount for loans classified as higher-priced. This is divided by total mortgage loan volume in the county-year:

$$\text{hp_share}_{ct} = \frac{\sum_i \text{LoanAmount}_{ict} \times \mathbf{1}[\text{HigherPriced}_{ict}]}{\sum_i \text{LoanAmount}_{ict}}.$$

Further, to examine whether the pricing of higher-priced mortgages differs according to whether the loan is purchased by another institution in the secondary market, I also construct a county-year measure of rate spreads for purchased loans using the loan-level HMDA data. Section A describes purchased loans in detail. The measure is

$$\text{avg_rate_spread_purchased}_{ct} = \frac{\sum_i \text{LoanAmount}_{ict} \times \text{RateSpread}_{ict} \times \text{Purchased}_{ict}}{\sum_i \text{LoanAmount}_{ict} \times \text{Purchased}_{ict}},$$

where i indexes individual higher-priced mortgage loans, RateSpread_{ict} is the loan's reported rate spread, and Purchased_{ict} is an indicator equal to one when the loan is classified as purchased.

Loan-to-Income Ratios

Loan-to-income (LTI) ratios are constructed from the loan-level HMDA data for each county and year. For loan i in county c and year t , the LTI ratio is calculated as

$$\text{LTI}_{ict} = \frac{\text{LoanAmount}_{ict}}{\text{Income}_{ict}}.$$

Loans with missing or non-positive income are excluded from the calculation. I also exclude observations with non-positive LTI ratios and outliers. Using this, the main county-year measure is constructed: avg_lti_{ct} is the arithmetic mean of the loan-level LTI ratios,

$$\text{avg_lti}_{ct} = \frac{1}{N_{ct}} \sum_i \text{LTI}_{ict},$$

where c denotes county, t denotes year, and i indexes individual mortgage loans. N_{ct} is the number of loans with a valid LTI ratio in county c and year t . Unlike the rate spread measures, the average LTI is not weighted by loan amount. For average LTI, each mortgage receives equal weight in the county-year average. This is intended, because it allows one to see the leverage of the typical borrower, rather than the leverage associated with a typical dollar of mortgage

lending.

Further, to examine whether changes in average leverage differ across lender types, I construct separate county-year averages for bank and non-bank loans following the same method as above. In a similar fashion, I also construct separate average LTI measures by mortgage product. Average LTIs for conforming, non-conforming and FHA loans are calculated as:

$$\text{avg_lti}_{ct}^p = \frac{1}{N_{ct}^p} \sum_{i \in p} \text{LTI}_{ict},$$

where p denotes the mortgage product category such as conforming, non-conforming, or FHA, and N_{ct}^p is the number of valid loans belonging to that category in county c and year t .

In sum, the resulting dataset contains one observation per county-year for the overall average LTI, bank average LTI, non-bank average LTI, and the product-specific average LTI measures. The same loan-level LTI construction is used throughout the analysis to ensure comparability across lender types and mortgage products.

To deal with the influence of outliers, I winsorize five variables at the county-year level. For values that are extreme on both tails, observations below the 1st percentile and above the 99th percentile are winsorized. This procedure is applied to `log_price_to_income`, non-bank average rate spread, and bank average rate spread. For values that are extreme on only the upper tail, observations above the 99th percentile are winsorized, while the lower tail is left unchanged. This procedure is applied to average rate spread and average rate spread purchased.

I calculate minority share as:

$$\text{Minority Share}_{ct} = 1 - \frac{\text{Population}_{ct}^{\text{White, NonHispanic}}}{\text{Population}_{ct}^{\text{Total}}}.$$

The data provide population counts by race and Hispanic origin, along with age and sex. I first aggregate the population across age and sex within each county-year-race/Hispanic-origin group. I then use these totals to identify the non-Hispanic White population and calculate the minority share. This approach uses the race and Hispanic-origin information together, so individuals who are Hispanic are classified consistently with their reported race and Hispanic origin.

For heterogeneity analysis, the county's baseline minority share measure is used rather than its contemporaneous value. I use baseline minority share measured in 2008, and therefore this is fixed for each county throughout the estimation period. This ensures that the heterogeneity measure captures counties' pre-existing demographic composition rather than changes that occur during the treatment period. Further, the baseline measure is used to distinguish counties with relatively high and low pre-existing minority populations in the heterogeneity analysis.

As an alternative measure, I also construct a narrower minority-share variable based on the Black or Hispanic population. This measure defines the minority population as individuals who

are either Black or Hispanic:

$$\text{MinShare}_{ct}^{\text{BH}} = \frac{\text{Population}_{ct}^{\text{Black} \cup \text{Hispanic}}}{\text{Population}_{ct}^{\text{Total}}}.$$

Individuals who are both Black and Hispanic are counted only once. This alternative measure is used to examine whether the heterogeneity results are sensitive to the definition of minority population.

C Empirical Strategy

Table 16. Exclusion Restriction: Reduced-Form Effect on Bank Lending Standards

	(1) Bank High-LTI Share	(2) Bank Average LTI
Regulatory Exposure Index $\times$ Post	0.0153*** (0.0025)	0.0766*** (0.0129)
Observations	23,130	23,130
F-statistic	38.48	35.09

Notes: Reduced-form regressions of bank-specific loan-to-income measures on regulatory exposure, testing whether banks themselves tightened lending standards in high-exposure counties, a channel that would bypass non-bank share and threaten the exclusion restriction (Section 5.4.2). Standard errors, clustered at the county level, are in parentheses. F-statistic tests the coefficient on exposure_post via `test`, not the joint model F. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 17. Monotonicity: First-Stage Effect by REI Tercile

	Tercile Effects
Tercile 2 $\times$ Post	0.0133*** (0.0038)
Tercile 3 $\times$ Post	0.0309*** (0.0040)
Observations	23,130
Clusters	2,378
F-statistic (T2 = T3)	22.40

Notes: Estimates from equation (5.1), replacing the continuous Regulatory Exposure Index with tercile indicators. Tercile 1 (lowest REI, omitted category) is the reference group. Coefficients show the additional effect on non-bank share for counties in Tercile 2 or Tercile 3, relative to Tercile 1. Standard errors, clustered at the county level, are in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 17 splits the cross-sectional distribution of the REI into three groups, with cutpoints at the 33rd and 67th percentiles. The results remain consistent with this tercile-split specification, as discussed in Section 5.4.4. T1 (low exposure) has REI ranging from 0 to 0.175 (mean 0.101); T2 (medium) from 0.175 to 0.335 (mean 0.254); T3 (high) from 0.335 to 1.0 (mean 0.471). Relative to T1, medium-exposure counties experienced 1.33 percentage points more non-bank share growth post-2013 (3.4% of the sample mean), and high-exposure counties experienced 3.09 percentage points more (7.8% of the sample mean).²⁸

Control Variables

Beyond the county and year fixed effects described in Section 5.2.1, all specifications include a vector of time-varying county-level controls X'_{ct} to absorb observable sources of variation in non-bank mortgage share and the outcome variables. The control set varies across outcomes because each local housing market or credit-condition outcome requires a different set of controls.

²⁸T3's range (0.335–1.0) is roughly three times wider than T1's or T2's (~ 0.17 each). This reflects the right-skewed REI distribution documented in Figure 2, where a long upper tail stretches the top tercile.

Table 18. Event Study: Non-Bank Share by Year Relative to 2012

	(1) Non-bank Share
Exposure $\times$ 2009	−0.0366*** (0.0119)
Exposure $\times$ 2010	−0.0170* (0.0095)
Exposure $\times$ 2011	−0.0217*** (0.0078)
Exposure $\times$ 2013	0.0262*** (0.0070)
Exposure $\times$ 2014	0.0720*** (0.0107)
Exposure $\times$ 2015	0.0574*** (0.0114)
Exposure $\times$ 2016	0.0370*** (0.0122)
Exposure $\times$ 2017	0.0648*** (0.0136)
Exposure $\times$ 2018	0.0681*** (0.0132)
Observations	23,130
Joint pre-trend F-statistic	4.37
Joint pre-trend p-value	0.0045

Notes: Coefficients from regressing non-bank mortgage share on the Regulatory Exposure Index interacted with year indicators; 2012 is the omitted reference year. County and year fixed effects are absorbed, and the control set described in Appendix C is included. Standard errors, clustered at the county level, are in parentheses. The joint pre-trend F-statistic tests whether the three pre-period coefficients (2009–2011) are jointly equal to zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In addition, variables that enter the construction of an outcome variable are excluded, since they would be endogenous to the dependent variable by construction.

The main control set, used for all local housing market outcome regressions, includes seven variables. *Lagged average borrower LTI* measures pre-existing leverage conditions in the county mortgage market, specifically last year's average loan-to-income ratio among mortgage borrowers. Counties where borrowers were already highly leveraged before non-bank expansion may have experienced different housing market dynamics independently of non-bank expansion; more leveraged borrowers are more reactive to price changes, which affects both price volatility and price growth. Including lagged LTI absorbs this pre-existing leverage condition. The lagged value is used because present LTI itself may be affected by current non-bank share.

Unemployment rate measures local labour market conditions in county c in year t . First, higher unemployment reduces mortgage demand and housing activity independently of non-bank expansion. Second, counties with weaker labour markets may also have had weaker banks in 2008, creating a potential correlation between the instrument and local economic conditions.

Lagged log house price index captures pre-existing house price levels in the county. House prices are persistent, so last year's house price would predict this year's price level. Without this control, the instrument could pick up counties already on high price trajectories rather than the effect of non-bank expansion.

Population growth captures demand-side pressure on the housing market. It directly increases housing demand and puts upward pressure on prices independently of credit supply.

Median household income captures local income conditions. It directly determines mortgage demand, borrowing capacity, and housing affordability independently of non-bank expansion. As higher income is a sign of creditworthiness, counties dominated by higher-income borrowers may attract more non-bank lenders.

County minority population share captures the demographic composition of the county. Non-bank lenders may expand more readily in counties with a particular demographic composition for reasons unrelated to Basel III regulatory pressure on banks. This control is excluded from the heterogeneity regressions, since baseline minority share enters the interaction term there directly.

Log level of residential building permits in the previous year captures housing supply conditions in the county. Counties with higher permits see active construction, and this absorbs credit supply shocks operating through new housing rather than price appreciation; without this control, the price results could reflect supply constraints. The lagged level is used because permits approved in a given year typically translate into housing units over the following 12 to 24 months and then feed into local housing demand and prices.

Poverty rate captures the lower tail of the local income distribution. Since median household income alone does not fully reflect affordability constraints faced by lower-income households, I control for the poverty rate. This is added only to the affordability control set and excluded for housing market and credit condition outcomes, since it risks introducing multicollinearity there.

All regressions include county and year fixed effects. Standard errors are clustered at the county

level.²⁹ The sample covers 23,130 county-year observations across 2,378 US counties over the period 2009–2018.

Additional Estimation Equations

$$\begin{aligned} \text{Nonbank_share}_{ct} = & \pi_1(\text{REI}_c \times \text{Post}_t) + \pi_2(\text{REI}_c \times \text{Post}_t \times \text{BaselineMinShare}_c) \\ & + X'_{ct}\gamma_1 + \delta_c + \delta_t + \nu_{ct}, \end{aligned} \quad (6)$$

$$\begin{aligned} (\text{Nonbank_share} \times \text{BaselineMinShare})_{ct} = & \pi_3(\text{REI}_c \times \text{Post}_t) + \pi_4(\text{REI}_c \times \text{Post}_t \times \text{BaselineMinShare}_c) \\ & + X'_{ct}\gamma_2 + \delta_c + \delta_t + \mu_{ct}, \end{aligned} \quad (7)$$

$$\begin{aligned} Y_{ct} = & \beta_1 \widehat{\text{Nonbank_share}}_{ct} + \beta_2 (\widehat{\text{Nonbank_share}} \times \widehat{\text{BaselineMinShare}})_{ct} \\ & + X'_{ct}\gamma + \delta_c + \delta_t + \varepsilon_{ct}. \end{aligned} \quad (8)$$

where $\text{BaselineMinShare}_c$ is minority population share in 2008, predetermined with respect to the post-2013 period.³⁰ $\text{REI}_c \times \text{Post}_t$ is the baseline regulatory exposure measure, and its interaction with $\text{BaselineMinShare}_c$ allows the first-stage relationship to vary with pre-existing minority shares. The coefficient π_2 therefore captures whether the increase in non-bank mortgage share induced by Basel III is larger or smaller in counties with higher baseline minority shares.

The coefficient β_1 captures the effect of non-bank share on the specified outcome variable at zero minority share, and β_2 captures how this effect changes as minority share rises. The total marginal effect at any given minority share level is $\beta_1 + \beta_2 \times \text{BaselineMinShare}_c$.

²⁹Clustering at the county level allows for correlation in the error term across years within the same county, a natural concern in panel data.

³⁰See Section 4.2.4 for the construction of baseline minority share.

D Additional Results

Table 19. House Price Growth, One-Year-Ahead

	(1) HP Growth (t+1)
Non-bank mortgage share	0.580*** (0.093)
Observations	20,820
F-statistic	46.64
Sample mean	0.0134
1 SD effect (pp)	9.284

Notes: Estimates from equation (5.2), instrumented by the REI interacted with Post (Section 5.3). Dependent variable is annual log house price growth measured one year after non-bank share, using the F. (forward) operator on the declared county-year panel. This specification loses each county’s final observed year relative to the contemporaneous specification (Table 5), reducing the sample from 23,130 to 20,820 observations. All specifications include county and year fixed effects and the control set described in Appendix C. Standard errors, clustered at the county level, are in parentheses. F-statistic is the Kleibergen–Paap rk Wald F-statistic on the excluded instrument. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 20. Additional Housing Stability Measures: Drawdown and Persistent Growth

	(1) Drawdown	(2) Persistent Growth
Non-bank mortgage share	−1.089*** (0.172)	4.030*** (0.695)
Observations	23,130	23,130
F-statistic	48.96	48.96
Sample mean	0.0573	0.5614

Notes: Estimates from equation (5.2), instrumented by the Regulatory Exposure Index interacted with Post (Section 5.3). Drawdown measures the proportional decline from a county’s six-year peak house price index (Appendix Section B). Persistent growth measures the three-year rolling share of years with positive house price growth (Appendix Section B). All specifications include county and year fixed effects and the control set described in Appendix C. Standard errors, clustered at the county level, are in parentheses. F-statistic is the Kleibergen–Paap rk Wald F-statistic on the excluded instrument. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Mechanisms: Reduced Form Equations

Table 21. Lending Volume by Loan Type: Conventional and FHA

	(1) Bank Conv.	(2) NB Conv.	(3) Bank FHA	(4) NB FHA
Regulatory Exposure Index $\times$ Post	-0.093*** (0.032)	0.264*** (0.058)	-0.758*** (0.160)	0.115* (0.070)
Observations	23,130	23,130	23,130	23,130
Exact % change	-8.9	30.3	-53.2	12.2

Notes: Reduced-form estimates of the effect of regulatory exposure on log lending volume, by lender type and loan type. Conventional loans include both conforming and non-conforming loans. All specifications include county and year fixed effects and the control set described in Appendix C. Standard errors, clustered at the county level, are in parentheses. Exact percent change is computed as $100(e^{\hat{\beta}} - 1)$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 22. Lending Volume by Loan Type: Conforming and Non-Conforming

	(1) Bank Conf.	(2) NB Conf.	(3) Bank Non-conf.	(4) NB Non-conf.
Regulatory Exposure Index $\times$ Post	-0.251*** (0.047)	0.250** (0.115)	0.210*** (0.041)	0.404*** (0.146)
Observations	23,130	23,130	23,130	23,130
Exact % change	-22.2	28.4	23.4	49.8

Notes: Reduced-form estimates of the effect of regulatory exposure on log lending volume, by lender type and conforming status. Conforming status requires a conventional loan to meet GSE underwriting criteria and fall below the applicable county conforming loan limit (Section 2.3). All specifications include county and year fixed effects and the control set described in Appendix C. Standard errors, clustered at the county level, are in parentheses. Exact percent change is computed as $100(e^{\hat{\beta}} - 1)$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 23. Mechanism Evidence: Bank Capital Response and Credit Reallocation

	(1) Bank Tier 1 Ratio	(2) Log Bank Volume	(3) Log Non-bank Volume
Post $\times$ Weak Bank (2008)	0.014*** (0.003)		
Regulatory Exposure Index $\times$ Post		-0.125*** (0.030)	0.161*** (0.037)
Observations	35,072	23,130	23,130

Notes: This table presents illustrative reduced-form evidence for the capital-response mechanism proposed in Section 5.2.2, and is not itself instrumented. The sample includes all HMDA-matched banks with available Call Report data, not only the subset of weak banks contributing to the Regulatory Exposure Index. This estimate does not distinguish a Basel III-specific capital response from general post-crisis capital rebuilding; see Section 5.4.2 for discussion of this identification concern. Standard errors are clustered at the bank level (Column 1) or county level (Columns 2–3), in parentheses. Exact percent change is computed as $100(e^{\hat{\beta}} - 1)$, not reported for Column (1) since the outcome is not in logs. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 24. Decomposition: Price and Income Components of Affordability

	(1) Log HPI	(2) Log Income
Non-bank mortgage share	0.896*** (0.141)	0.236*** (0.065)
Observations	23,149	23,130
F-statistic	43.94	42.89
% change (1 SD)	15.4	3.9

Notes: Estimates from equation (5.2), instrumented by the Regulatory Exposure Index interacted with Post (Section 5.3). Column (1) reports the effect on the log level of house prices; Column (2) reports the effect on log median household income. Median household income is excluded from the control set in Column (2) since it is the outcome. All specifications include county and year fixed effects. Standard errors, clustered at the county level, are in parentheses. F-statistic is the Kleibergen–Paap rk Wald F-statistic on the excluded instrument. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 25. Baseline (2008) Characteristics by Minority Share

	Low-Minority	High-Minority
Regulatory Exposure Index	0.278	0.275
Non-bank mortgage share	0.287	0.348
Bank mortgage share	0.713	0.652
FHA share	0.176	0.217
Counties	1,184	1,194

Notes: Cross-sectional comparison of 2008 county characteristics, split at the sample median of baseline minority population share, restricted to counties appearing in the main estimation sample. Low-minority counties are at or below the median; high-minority counties are above the median. Non-bank and bank mortgage shares and FHA share are volume-weighted; all figures are unweighted county averages within each group.

F Additional Robustness Checks

Average credit rate spread with local lender concentration (HHI). Local lender concentration could affect credit rate spreads independently of non-bank expansion: counties with the same non-bank share may have very different mortgage markets, with lending spread across many lenders in one county and concentrated among a few in another. As a robustness check, I add the lagged Herfindahl–Hirschman Index (HHI) of mortgage lending as a control.

The HHI is calculated at the county-year level using lenders’ shares of total mortgage lending, with shares weighted by loan volume. A higher HHI indicates that lending is concentrated among fewer lenders, while a lower HHI indicates a more fragmented market. I use the lagged HHI to avoid controlling for changes in concentration that may themselves result from the non-bank reallocation.

Table 26 shows that including lagged HHI increases the estimated effect of non-bank share across all three average credit rate spread measures. For the total rate spread, the coefficient increases by 0.616, equivalent to a 22.2% increase relative to the sample mean, compared with 18.1% without the concentration control. For the non-bank spread, the coefficient increases by 1.516, or 56.2% of the sample mean, compared with 45.8%. For the bank spread, it increases by 0.780, or 21.4% of the sample mean, compared with 16.2%.

The results suggest that the average rate spreads are not driven by differences in local mortgage market concentration; if anything, accounting for concentration strengthens the estimated relationship between non-bank share and credit rate spreads.

Table 26. Robustness: Rate Spread with Local Market Concentration

	(1) Total	(2) Non-bank	(3) Bank
Non-bank mortgage share	3.294*** (0.871)	8.221*** (1.889)	3.232*** (0.892)
Observations	23,047	22,254	22,702
F-statistic	29.93	29.53	28.66
% change (1 SD)	22.2	56.2	21.4

Notes: Estimates from equation (5.2), instrumented by the Regulatory Exposure Index interacted with Post (Section 5.3), with lagged Herfindahl–Hirschman Index (HHI) of local lender concentration added as an additional control, measured in the prior year to avoid capturing concentration changes that are themselves a consequence of the reallocation under study. HHI is constructed across all HMDA-reporting lenders, not banks alone, and is therefore not fully independent of non-bank share by construction. All specifications include county and year fixed effects and the control set described in Appendix C. Standard errors, clustered at the county level, are in parentheses. F-statistic is the Kleibergen–Paap rk Wald F-statistic on the excluded instrument. % change (1 SD) is the percentage change in the outcome from a one standard deviation increase in non-bank share, relative to each column’s own sample mean. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Outcome Measurement

As a robustness check, house price volatility is re-measured using a five-year rolling window instead of the three-year window used in the main specification. Table 27 shows that a one percentage point increase in non-bank share reduces five-year house price volatility by 0.058 percentage points, compared with 0.055 percentage points using the three-year measure. The

coefficient is statistically significant at the 10% level ($p = 0.053$), although it is estimated less precisely than in the main specification.³¹ This result suggests that the volatility finding is not driven by the choice of the three-year window.

Table 27. Robustness: House Price Volatility, Five-Year Window

	(1) HPV (5yr)
Non-bank mortgage share	−0.0585* (0.0302)
Observations	23,130
F-statistic	48.96
Sample mean	0.0375
1 SD effect (pp)	−0.936

Notes: Estimates from equation (5.2), instrumented by the Regulatory Exposure Index interacted with Post (Section 5.3). Dependent variable is house price volatility measured over a five-year rolling window instead of the main specification’s three-year window (Table 5, Column 1). All specifications include county and year fixed effects and the control set described in Appendix C. Standard errors, clustered at the county level, are in parentheses. F-statistic is the Kleibergen–Paap rk Wald F-statistic on the excluded instrument. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 28. Robustness: Heterogeneity by Baseline Black/Hispanic Share – First Stage, Housing Outcomes, and Affordability

	(1) First Stage	(2) HPV (3yr)	(3) HP Growth	(4) 3yr Cum. Growth	(5) Log Price-to-Income
REI × Post	0.059*** (0.011)				
REI × Post × Baseline Black/Hispanic Share	0.138*** (0.029)				
Non-bank share		−0.067** (0.028)	0.813*** (0.121)	2.055*** (0.301)	0.719*** (0.120)
Non-bank share × Baseline Black/Hispanic Share		−0.097*** (0.020)	0.120 (0.074)	0.388** (0.184)	0.315*** (0.077)
Observations	23,130	23,130	23,130	23,130	23,130
F-statistic	28.09 / 22.39	29.56	29.56	29.56	27.57

Notes: As a robustness check on Table 11, baseline minority share is redefined using Census Black and Hispanic population share specifically, rather than the broader non-white population share used in the main specification. Column (1) reports the first stage. Columns (2)–(5) report the corresponding IV estimates, instrumented by this interaction system. Median household income is excluded from the control set in Column (5), since it enters the affordability outcome’s construction directly. All specifications include county and year fixed effects and the control set described in Appendix C, excluding minority share as a standalone control since it enters the interaction term. Standard errors, clustered at the county level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

³¹The Kleibergen–Paap F-statistic remains essentially unchanged, at 48.96.

Statement of Authorship

I hereby confirm that the work presented has been performed and interpreted solely by myself except for where I explicitly identified the contrary. I assure that this work has not been presented in any other form for the fulfillment of any other degree or qualification. Ideas taken from other works in letter and in spirit are identified in every single case.

Bonn,

Chandra Maharshi Valasa